Certifying the Judge — Complete Notes

Definition of a Trustworthy AI Judge, and a Certification Protocol

11 August 2026

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Certifying the Judge — Complete Notes

Definition of a Trustworthy AI Judge, and a Certification Protocol

Compiled 17 August 2026 from thirty-one photographed handwritten pages: loose sheets, notebook pages, an annotated printout of *Certifying the Judge* (Figs. 13–14), a wall-pinned sheet, one page written on a plane (`38700.jpg`) containing the proposed formal definition, a follow-up page (`38704.jpg`) disambiguating the method indices and raising the h_j-pointer problem, a work-in-progress page (`38707.jpg`) instantiating the method with real VeriBench numbers, two effective-parameter pages (`39296`/`39297`) separating $b' \subseteq b$ from b, a naming page (`20260812_1324.jpg`) rejecting the name "Trust Index", and a ratification page (`20260812_1348.jpg`) fixing the final notation system: TW_m / TI_m / PTI_p.

Living document. This is a work in progress and will be revised as further pages arrive. Everything captured so far is included, including material that will not survive into a paper — the stream-of-thought reasoning is retained deliberately, because the discarded branches are what justify the surviving ones.

How to read this. Part I is the *definition* — what trustworthiness **is**, stated in terms of quantities that require humans. Part II is why that definition cannot be checked directly at scale. Part III is the *certificate* — the label-free instrument. Part IV is the *protocol*, written algorithmically. Part V is the logical bridge between certificate and definition, which is the only thing the paper actually has to establish. Parts VI–VII are threats and open items. Appendices carry the verbatim transcriptions and a notation-reconciliation log.

Uncertain readings from the handwriting are marked `[?]`.

Part 0 — Notation, with types

The handwritten pages use several notations for the same object. Everything below is standardized; Appendix C logs every place the originals differ and what was chosen.

0.1 Sets and objects

Symbol	Type	Meaning
$\mathcal{X}$	set	Task inputs (problem statements, spec prompts)
$\mathcal{Y}$	set	Candidate artifacts (e.g. Lean 4 files, code)
$\mathcal{S}$	set	Score range, w.l.o.g. $[0, 1]$
$b \in B$	set of items	Benchmark. $b = \{(x_i, y_i^*)\}_{i=1}^{n_b}$. VB = VeriBench
$a \in A$	$\mathcal{X} \to \mathcal{Y}$	Agent (a *solver* — usually an AI agent, e.g. Codex-class)
$L_A \subseteq A$	subset	Leaderboard agents — the subset we actually must score
$\hat{y} = a(x)$	$\mathcal{Y}$	Agent's response: the artifact being judged
y^*	$\mathcal{Y}$	Human/gold reference. **Optional** in the conditioning. **Indexed by b, never by the judge**
r	prompt	Rubric
$\pi \in \Pi$	persona	Judge persona (strict / lenient / risk-taking / conservative …)
$aj \in AJ$	$\mathcal{X} \times \mathcal{Y}^* \times \mathcal{Y} \to \Delta(\mathcal{S})$	**AI judge.** Identity is the pair (r, π)
$h_j \in H(b)$	$\mathcal{X} \times \mathcal{Y}^* \times \mathcal{Y} \to \Delta(\mathcal{S})$	**Human judge.** Identity is the pair (r, π) — a person, or a described persona

0.2 Distributions and scores

Symbol	Type	Meaning
$P_\theta(s \mid x, y^*, \hat{y}, aj)$	$\Delta(\mathcal{S})$	AI judge's score distribution. θ = LLM weights (open or closed)
$P_H(s \mid x, y^*, \hat{y}, h_j)$	$\Delta(\mathcal{S})$	Human judge's score distribution
$\hat{s} \sim P_\theta(\cdot \mid x, y^*, \hat{y}, aj)$	$\mathcal{S}$	A sampled AI score
$s^* \sim P_H(\cdot \mid x, y^*, \hat{y}, h_j)$	$\mathcal{S}$	A sampled human score
$S_{aj} = \{\hat{s}_i\}_{i=1}^n$	vector in $\mathcal{S}^n$	AI scores **indexed by item** i
$S_{h_j} = \{s_i^*\}_{i=1}^n$	vector in $\mathcal{S}^n$	Human scores **indexed by item** i

0.3 Operators and indices

Symbol	Type	Meaning
$aj \approx_\Pi h_j$	relation on $AJ \times H$	**Persona match.** The two judges are asked to be the same kind of rater
$d_M(\cdot, \cdot)$	$\Delta(\mathcal{S})^2 \to \mathbb{R}_{\geq 0}$	**Matching criterion** M — the sense in which two judges "agree"
$\mathcal{M}$	(conditioning set) $\to AJ$	**Judge-synthesis operator**: the method that *produces* a judge
P_k	$AJ \to [0, 1]$	Behavioural property k, higher is better. **Judge-indexed**
PTI_p	$AJ \to [0, 1]$	**Trust Index.** Label-free. **Judge-indexed**
TI_{m_1}	$AJ \to [-1, 1]$	Human alignment of a judge. Label-requiring. **Judge-indexed**
GM	$[0, 1]^k \to [0, 1]$	Geometric mean

> **[!] The single most important type fact in these notes.** TI and TI_{m_1} are both **functions of a judge**, i.e. one scalar per judge. Human *scores* are functions of an **item**. You never correlate TI against human scores. You compute $\text{TI}_{m_1}(aj)$ by correlating **over items** for a fixed judge, then correlate TI against TI_{m_1} **over judges**. Two nested correlations, two different index sets. Getting this wrong is a type error, and it is the fastest way to be desk-rejected.

0.4 Naming decisions (FINAL — ratified on `20260812_1348.jpg`)

The naming went through three stages, each rejected for a stated reason. The evolution is kept because the rejections carry design information.

Stage	Name for the label-free scalar	Rejected because
1	Trust Index (PTI_p)	Collides with TW_k; and the name asserts the contested claim (a "Trust Index = 0.9" reads as "90% trustworthy" before any bridge exists — polygraph named as lie detector)

Stage	Name for the label-free scalar	Rejected because
2	Proxy Trust Index	`CI` collides with confidence interval; `CC` (Certification Coefficient) collides with Claude Code; and — the author's own objection — TI_m *also* certifies the judge, arguably better, so "certification" does not distinguish the instrument
3	**Proxy Trust Index,** PTI_p	**Adopted.** The relationship to the gold scalar is in the name: PTI_p is a *scalable-oversight approximation to* TI_m

The ratified three-slot system

Slot	Object	Name	Notation	Measurement-theory role	Needs humans?
1	The **predicate** — the definition	Trustworthiness	$\mathrm{TW}_m(aj)$	**Construct**	—
2	The **gold scalar** — how well TW_m holds, measured on a specified panel/items	Trust Index	$\mathrm{TI}_m(aj)$	**Criterion**	**Yes**
3	The **proxy scalar** — label-free approximation to slot 2	Proxy Trust Index	$\mathrm{PTI}_p(aj)$	**Instrument / surrogate**	**No**

The third column matters strategically: construct / criterion / instrument is the vocabulary of Salaudeen et al. 2025 ("Measurement to Meaning," the validity-centered evaluation framework) — **Sanmi's own lab, already cited in the workshop paper's related work.** Adopting these role labels costs nothing, hard-aligns the paper with the advisor's framework, and gives reviewers a citation-anchored home for the construct/criterion gap (TI_m is *not* "true trustworthiness" either — it is a sampled, scope-dependent criterion affected by panel, persona, items, agents, rubric, and aggregation). In prose, prefer "surrogate" over "proxy" as the technical term (FDA surrogate-endpoint usage is the exact analogy: a substitute whose predictive relationship to the outcome must be validated, and which can fail outside its validated context); PTI keeps its name for continuity with the ratified handwriting.

Subscript discipline, following the L_p-space pattern (capital letter names the family, subscript indexes the variant):

- m indexes the **target side**: which matching criterion defines trustworthiness. Ratified ladder (lowercase, three levels): m_0 distributional; m_1 correlational ("most common"); \$m_2 = m_1 + \$ stability ("fav"). Written TW_2 or TW_{m_2}.
- p indexes the **proxy side**: which necessary conditions are probed. $p \subseteq \{1, \ldots, K\}$ is an **index set**, default $p = \{1, 2, 3, 4\}$, so ablations are free notation: $\mathrm{PTI}_{p\setminus\{4\}}$. p **is locked before the bridge** — choosing $\hat{p} = \arg\max_p \mathrm{Corr}(\mathrm{PTI}_p, \mathrm{TI}_m)$ and reporting that correlation is model selection over $2^K - 1$ subsets and reintroduces the circularity.

- The paper's empirical object is the **pair** (p, m): for which pairs does $\text{PTI}_p \approx_C \text{TI}_m$ across judges?

$$\text{PTI}_p(aj) \;=\; \text{GM}(P_k(aj) : k \in p) \qquad\qquad \text{TI}_{m_1}(aj) \;=\; \text{Corr}_i(s_{aj}(i),\, s_{h_j}(i))$$

TI_{m_1} is the object earlier drafts called TI_{m_1}; PTI_p is the object earlier drafts called PTI_p and briefly PTI_p. Older sections retained below use the old symbols with this mapping; the *paper* uses only the ratified names, renamed in one pass.

The asymmetry, stated correctly (incorporating the author's objection)

Both scalars are empirical — the author is right that neither guarantees anything at finite data. The reject-only asymmetry lives one level up, in the **estimand**:

Error source	TI_m	PTI_p
Construct gap (is TW_m the right formalization?)	shared	shared
Estimand gap (at infinite data, does the metric equal the TW_m scalar?)	**zero by construction** — TI_m *is* the scalar realization of TW_m	**nonzero forever** — necessity $\neq$ sufficiency; infinite probes never close it
Estimation error (finite labels / probes)	nonzero	nonzero

So: PTI_p's acceptance failure is *structural*; TI_m's is merely *statistical*. The ratification page concedes both halves in its own margins — rejection is annotated "approximately", and acceptance is written "certifies" **in scare quotes**, conditional on the demonstrated bridge ("cuz we demonstrated that PTI **does** correlate with TI — key!"). That is the licensed-acceptance semantics of §3.4, in the author's hand.

> Abbreviation rules: never `CI` (confidence interval), never `CC` (Claude Code), never a hat ($\widehat{\text{TI}}$ — a hat claims estimator semantics, and PTI_p does not converge to TI_m under any amount of probing).

Part I — The definition

1.1 Prose statement (from `38700.jpg`, cleaned)

> **Definition (proposed) — Trustworthy AI Judge.** For all benchmarks $b \in B$, the AI judge should work for **all agents**, or for the subset of agents on the leaderboard ($\forall a \in A$, or $a \in L_A$, $L_A \subseteq A$). If the AI judge **matches in persona** (e.g. strict judge / nice judge / non-strict, etc.) the human judge — i.e. if $aj \approx_\Pi h_j$ — then the two judges should **match** when they judge/ score how well agent a did on solving tasks from benchmark b. (E.g. the task is: given question/data point x, produce $\hat{y} = a(x)$ such that $\hat{y} \approx y^*$, or just a good $\hat{y}$.)

1.2 Formal statement

Definition 1 ((M, ε)-trustworthy AI judge). An AI judge aj is (M, ε)-**trustworthy** with respect to benchmark family B, agent set A (or L_A), and human judge population H iff

$$\forall b \in B,\;\; \forall a \in A \text{ (or } L_A),\;\; \forall (x, y^*) \in b,\;\; \forall h_j \in H(b) \text{ with } aj \approx_\Pi h_j :$$

$$\boxed{d_M\Big(\ P_\theta(\cdot \mid x,\, y^*,\, \hat{y}{=}a(x),\, aj),\ \ P_H(\cdot \mid x,\, y^*,\, \hat{y}{=}a(x),\, h_j)\ \Big)\ \leq\ \varepsilon}$$

Three things to notice, each of which was written explicitly in the margins:

1. y^* **is optional.** Dropping it gives the *reference-free* judge; keeping it gives the *reference-based* judge. Both are legitimate; the definition is parameterized by which.
2. **The persona precondition $aj \approx_\Pi h_j$ is a genuine precondition, not decoration.** > "obviously if $h_j \equiv$ risk-taker persona but $aj \equiv$ conservative/shy, then the two > cannot be made to match. lol"

   Without it, the quantifier ranges over the full cross-product of AI judges and humans, and the definition is unsatisfiable whenever the human pool is heterogeneous. This is the structural explanation for the observed sign flips between diverse and narrow judge panels: a diverse panel violates the precondition by construction.
3. **The match is required per-item, over all agents and all benchmarks.** That is what makes the definition strong, and what makes it expensive (Part II).

1.3 The matching-criterion ladder

M is a parameter of the definition, not a fixed choice. Two handwritten pages give overlapping ladders (`38700.jpg` options 0/1/2; `37242.jpg` levels 0/1/2/3). Unified:

M	Name	Condition	Needs humans?	Note
M_0	**Distributional**	$P_\theta(\cdot) = P_H(\cdot)$ (or $\approx$ under KL / TV / total information)	Yes	*"Strongest / hardest: AI judge produces the exact distribution as the human person in question."*
M_1	**First moment**	$\mathbb{E}_{P_\theta}[s] \approx \mathbb{E}_{P_H}[s]$	Yes	On average the scores match
M_2	**First + second moment**	M_1 **and** $\mathrm{Var}_{P_\theta}[s] \approx \mathrm{Var}_{P_H}[s]$	Yes	Judge *reproduces* human disagreement
M_3	**Correlational**	$\mathrm{Corr}(S_{aj}, S_{h_j})$ high, over items	Yes	*"Most common"*. Corr $\in$ {Spearman, Kendall, Pearson, R^2, ICC}
M_4	**Correlational + stable**	M_3 **and** $\mathrm{Var}_{P_\theta}[s] \approx 0$	Corr term: yes. Stability term: **no**	*"My favourite modification to 1: add stability, e.g. aj is low variance/std."*

Written as on the page:

$$\underbrace{\mathrm{Corr}(S_{aj} \approx S_{h_j})}_{\text{needs humans}} \quad \textbf{and} \quad \underbrace{\mathrm{Var}_{P_\theta(s \mid x, y^*, a(x),\, aj)}[S_{aj}] \approx \text{low}}_{\text{label-free}}$$

with the correlational term abbreviated $\| S_{aj} \approx_{\text{corr}} S_{h_j} \|$.

[!] M_2 **and** M_4 **are in direct tension and cannot both be the target.** M_2 asks the judge to **reproduce** human disagreement; M_4 asks it to **eliminate** its own. One is a *fidelity* target, the other a *reliability* target. The Trust Index built in Part III uses the M_4 variance term $(1 - \text{std})$. **Pick** M_4**, say so once, and never claim** M_2 **anywhere in the same paper.** The self-annotation *"word didn't depend on humans lol…"* next to the low-variance condition is the observation that makes M_4 the right choice: its stability half is measurable with **zero annotation**, which is what makes the whole certification programme possible.

1.4 The judge-synthesis operator — this is what "scaling" means

`38700.jpg` introduces an object that does not appear on any other page, and it resolves an open question from the earlier notes (*"do we really need to scale it? what does scaling mean here?"*).

> "we want a method that produces such judges, e.g. $\mathcal{M}_0(B, A, h_j) = aj$, or $\mathcal{M}_1^s(b, a, h_j) = aj$, or $\mathcal{M}_2^s(b, A, h_j) = aj$" → *"want to condition on less"*

So there is a hierarchy of **judge-synthesis operators**, ordered by their **conditioning set**:

Operator	Conditions on	Produces	Generality
$\mathcal{M}(b, a, h_j)$	one benchmark, **one agent**, one persona	a judge valid for that (b, a) pair	weakest
$\mathcal{M}(b, A, h_j)$	one benchmark, **all agents**, one persona	a judge valid for benchmark b	middle
$\mathcal{M}(B, A, h_j)$	**all** benchmarks, **all** agents, one persona	a universally valid judge	strongest

Definition 2 (scaling). A certification method *scales along an axis* iff the judge-synthesis operator does **not** condition on that axis. The three scaling axes named in the earlier notes —

$$\forall a \in A, \qquad \forall b \in B, \qquad \forall aj \in AJ$$

— are exactly the three arguments you are trying to remove from $\mathcal{M}$'s conditioning set. *"Want to condition on less"* is the whole research programme in four words.

This is the answer to "what does scaling mean here?" It is not "run on more items." It is "remove an argument from the conditioning set of the operator that produces the judge."

Recommended scope. The handwritten pages answer their own scope question: > *"all benchmarks? Or just for VB? ↪ I think just for VB."*

So: fix $b = \text{VB}$, target $\mathcal{M}(\text{VB}, A, h_j)$ — generalize over agents and over judges, **not** over benchmarks. Claim exactly that and nothing more.

1.5 Disambiguating M_i — the definition index vs. the conditioning index

`38704.jpg` opens with exactly this as its stated next task:

> **TODO: Next task:** Clarify what M_i's mean, especially M_2, M_2', M_1 + what conditioning on h_j means.

The confusion is real and it is structural: **three different things are competing for one subscript.**

What the subscript is doing	Where it appears	Example
(i) which **definition of trustworthiness** the judge must satisfy	`38700` options 0/1/2; `38704` writes these as TW_i	TW_2 = correlational + stable
(ii) which **conditioning set** the method is allowed	`38700` $\mathcal{M}_0(B, A, h_j)$, $\mathcal{M}_1^s(b, a, h_j)$, $\mathcal{M}_2^s(b, A, h_j)$	(b, A, h_j) vs (B, A, h_j)
(iii) a **primed variant** of a conditioning set	`38704` M_2 vs M_2'	$A \to L_A$

`38704` already separates (i) from (ii) by naming the definition TW_i:

> M_i is a **method** that outputs a judge with the Def of trustworthiness that we want in (2) / Def TW_i (Trustworthy).

Recommendation: finish the job and stop numbering methods entirely. Write

$$\boxed{\mathcal{M}^{\mathrm{TW}_k}[\text{ conditioning set }] \longrightarrow aj}$$

- **Superscript** = which definition of trustworthiness the output is required to satisfy.
- **Bracket** = the conditioning set, written out in full. **No number.**

The conditioning set is the interesting object and it is short enough to write literally; a numeric alias for it buys nothing and costs exactly the confusion recorded in the TODO. Every method on both pages then becomes unambiguous:

Handwritten name	Unambiguous form	Reads as
$\mathcal{M}_0(B, A, h_j)$	$\mathcal{M}^{\mathrm{TW}_k}[B, A, h_j]$	one judge for **all** benchmarks and **all** agents
$\mathcal{M}_1^s(b, a, h_j)$	$\mathcal{M}^{\mathrm{TW}_k}[b, a, h_j]$	per-benchmark, **per-agent** — weakest
$\mathcal{M}_2^s(b, A, h_j)$	$\mathcal{M}^{\mathrm{TW}_k}[b, A, h_j]$	per-benchmark, **all** agents
$\mathcal{M}_{2'}(b, L_A, h_j)$	$\mathcal{M}^{\mathrm{TW}_2}[b, L_A, h_j]$	per-benchmark, **leaderboard** agents ← **the working target**

1.5.1 The working target, boxed on the page

$$\mathrm{Method}_{2'} \;=\; \mathcal{M}(b, L_A, h_j) \;=\; aj \qquad \longrightarrow \qquad \mathcal{M}^{\mathrm{TW}_2}[\,\mathrm{VB},\; L_A,\; h_j\,] \;=\; aj$$

> "So fix a benchmark, e.g. VB; fix a leaderboard $L_A \subseteq A$; and a human h_j (strict person, or person Rylan)."

In words: **fix VeriBench, fix the leaderboard agents, fix a human persona; produce an AI judge that correlates with that persona and is stable.** That is the whole claim, and it is a defensible one. The choice of TW_2 is justified on the page as *"since it's my favourite one … actually doable in practice, I think … + has nice conditions → matches/corr human in a stable way."*

TW_2 is the criterion called M_4 in the merged ladder of §1.3 (correlational + stable). The two numberings are recorded together in Appendix C so they can be cross-read.

> **[!]** $A \to L_A$ **is a weakening of Definition 1, not a notational convenience.** Restricting the agent quantifier from all agents to leaderboard agents makes the claim strictly easier and strictly narrower. It is the right call — leaderboard agents are the ones that actually need scoring — but it must be stated as a parameter of the definition and it must be the version claimed in the abstract. A definition quantified over A and an experiment run over L_A is the same species of gap that produced the desk rejection.

1.6 The h_j pointer problem

This is the sharpest new observation on `38704`, and it is a genuine hole in Definition 1.

> "assume we have h_j, e.g. a strict judge (e.g. Rylan) / persona. So the method could take this person and output a judge aj s.t. it has property Def TW_i … **But in practice we don't have a pointer like this one to the person/persona.** Instead we usually have scores/labels/judgments from that human judge / target judge(s) — e.g. **10 scores** according to the persona/person."

Definition 1 writes h_j as though it were an argument you can condition on: a pointer to a person, or to a persona. **Operationally that pointer does not exist.** What exists is a finite sample of that person's behaviour. So the honest signature of the method is not

$$\mathcal{M}^{\mathrm{TW}_2}[\,b,\, L_A,\, h_j\,] \qquad \text{but} \qquad \boxed{\mathcal{M}^{\mathrm{TW}_2}[\,b,\, L_A,\, D_{h_j}\,] \;=\; aj}$$

$$D_{h_j} \;=\; \{(x_i,\ \hat{y}_i,\ s_i)\}_{i=1}^{n}, \qquad s_i \sim P_H(\cdot \mid x_i,\, y_i^*,\, \hat{y}_i,\, h_j)$$

The persona is **latent**; D_{h_j} is the only evidence for it. Three consequences, all of which should be stated in the paper:

(a) Persona identifiability is a sample-complexity question, and it is open. How large must n be for $\mathcal{M}[b, L_A, D_{h_j}]$ to produce a judge that is TW_2-trustworthy with respect to the *true* h_j rather than the sample? The page writes $n \approx 10$. Ten ratings is a very small sample from which to recover a rater's scoring behaviour, and nothing currently bounds the error this introduces. This is a clean, statable open problem and a natural place for an actual theorem.

(b) Human labels enter in TWO places, not one. This changes the cost story and it must be said out loud, or a reviewer will find it:

Site	What labels buy	Scales with
Persona conditioning — D_{h_j} inside $\mathcal{M}$	Identifies the target persona so a judge can be built	number of **personas**, $O(n)$ each
Bridge validation — Algorithm 5	Establishes that TI predicts TI_{m_1}	number of **judges** in the validation pool, once

Neither scales with $|A|$ or with the number of candidate judges screened. **That** is the amortization claim, and it should be stated in exactly this form: cost is $O(1)$ in agents and in screened judges, $O(n)$ in personas. Not "no human labels" — that claim is false and easy to puncture.

(c) The method consumes labels; the certificate does not. $\mathcal{M}$ *builds* a judge using D_{h_j}. TI *checks* a judge using nothing. These are different budget lines and a reader will conflate them unless the paper separates them explicitly. It also means TI can be applied to judges that $\mathcal{M}$ did not produce — third-party judges, new frontier models — which is precisely the $\forall aj$ scaling axis.

1.6.1 Person vs. persona — the overload flagged on the page

> "or person + persona ≈ persona (**overloading terminology**)"

with the accompanying identity bracket:

```
model/agent }   prompt / θ /
identity    }   person
```

The overload is real. Resolve it by keeping the two components separate on both sides:

$$aj = (\underbrace{\theta}_{\text{model / weights}}, \underbrace{r \text{ containing } \pi}_{\text{prompt / rubric}}) \qquad h_j = (\underbrace{\text{person}}_{\text{individual index}}, \underbrace{\pi}_{\text{described persona}})$$

- **Person** = an individual human rater (Rylan, Steven, Jolene, Kelly, Simon). Idiosyncratic. **Not transferable and not reproducible.**
- **Persona** π = a *described role* (strict / lenient / risk-taking / conservative). Specifiable in a rubric, and therefore **the only component an AI judge can be asked to match.**

$\approx_\Pi$ is a relation on the **persona component only**. You cannot ask an AI judge to match a *person*; you can only ask it to match a *persona*.

> **This gives you a noise ceiling, and you should report it.** Whatever variance remains among people *within* the same persona is irreducible person-level idiosyncrasy. It upper-bounds any achievable $\text{TI}_{m_1}(aj)$. Estimate it as the mean inter-person correlation within a persona, call it $\rho_{\max}$, and report $\text{TI}_{m_1}(aj)/\rho_{\max}$ alongside the raw number. This is standard in the rater-agreement literature, it is cheap given data you already have, and it converts "our judge only gets $\rho = 0.6$" into "our judge reaches 90% of the human ceiling" — which is both more honest and considerably stronger.

1.6.2 Three possible targets — pick one and name it

> "or even a collection of scores from different humans. In the past all the humans would get the same prompt/rubric for judging and produce judgments, & we want an aj / Method that outputs a good TW_2 judge."

That sentence describes what the existing human data actually is: **one shared rubric, many people, pooled.** Which means the persona was *nominally* held fixed by the shared rubric while *actually* varying by person. Three distinct targets are now on the table:

Target	Definition matches	Reproducible?	Notes
T_1 **Individual person**	$P_H(\cdot \mid \cdots, \text{person})$	No	Matching one rater's idiosyncrasy. Not a scientific target
T_2 **Persona**	$P_H(\cdot \mid \cdots, \pi)$	Yes	What $\approx_\Pi$ is for. **Recommended**
T_3 **Panel consensus**	$\text{agg}_{h \in H}\, P_H(\cdot \mid \cdots, h)$ under a shared rubric	Yes	What the existing data supports. Requires **dropping** $\approx_\Pi$ from Definition 1 and replacing it with a stated aggregation operator

> **[!] This is very likely the mechanism behind the observed sign flips.** Pooling a heterogeneous panel under one shared rubric produces a T_3 target. A judge built and evaluated as though it were chasing T_2 is then being scored against something no persona-matched judge is trying to match. The identity-check signal going negative on a diverse panel and positive on a narrow panel is exactly what that mismatch predicts: a narrow panel is accidentally T_2-like, a diverse panel is irreducibly T_3. **This is checkable against data already collected** — persona-label the existing raters, recompute TI_{m_1} within persona strata, and see whether the flip disappears. If it does, the negative result becomes a positive one with a mechanism, which is a substantially better paper than a measurement-lessons paper.

1.6.3 The concrete instantiation (`38707.jpg`) — and what it exposes

`38707.jpg` instantiates $\mathcal{M}^{\mathrm{TW}_2}$ with actual values. This is the first page where the abstract machinery touches real data, and it immediately surfaces three problems.

$$\mathcal{M}^{\mathrm{TW}_2}\big[\ \underbrace{\mathrm{VB}_{v1}}_{\text{VeriBench v1}}\ ,\ \underbrace{L_A}_{\text{leaderboard}}\ ,\ \underbrace{D^S_{h_j}\ \text{or}\ D^S_H}_{\text{score sets}}\ \big]\ =\ aj$$

with, as written on the page,

$$L_A = \{\ \underbrace{\text{codex}}_{\text{GPT-5.6 ultra}}\ ,\ \underbrace{\text{CC}}_{\text{Fable 5 max}}\ ,\ [\text{illegible}]^{[?]},\ \text{Qwen},\ \text{Gemini}\}$$

$$D^S_{h_j} = \Big\{\, s \sim P_h(s \mid x,\, y^*,\, \hat{y} = a(x)) \,\Big\}^{N^S_{h_j}} \qquad D^S_H = \bigcup_{h_j \in H} \Big\{\, s \sim P_h(s \mid \cdots) \,\Big\}$$

$$H = \{\text{Ethan},\ \text{Yegor},\ \ldots\}, \qquad |D^S_H|\ \approx\ 75\ =\ 5\ \text{people} \times 15\ \text{judgments}$$

with marginal arithmetic $5\cdot 10,\ 5\cdot 5 = 5(10+5) = 5(15)$ — i.e. **the 15 per person looks like two batches of 10 and 5**, which usually means two annotation rounds and therefore a possibly non-uniform design.

Problem 1 — the union destroys the person index $D^S_H = \bigcup_{h_j \in H}\{\cdots\}$ is written as a **set union**. A union is a bag of scores with the rater identity thrown away. If the pipeline actually does this, then downstream you cannot:

- persona-stratify (§1.6.2) — the T_2 / T_3 question becomes unanswerable;
- estimate $\rho_{\max}$ (§1.6.1) — no within-persona inter-person correlation is computable;
- compute per-person TI_{m_1}, or check whether the sign flip is person-driven.

Fix: keep it an indexed family, not a union.

$$D^S_H\ =\ \{\,(h_j,\ x,\ \hat{y},\ s)\ :\ h_j \in H\,\} \qquad \text{not} \qquad \bigcup_{h_j \in H}\{s\}$$

This is a one-line notational change with real statistical consequences, and it should be checked against what the code actually stores, not just what the notation says.

Problem 2 — $n = 15$ makes the *inner* correlation underpowered too §Alg. 5 already flags that the **outer** correlation (over judges) is underpowered. This page reveals that the **inner** correlation may be as well. If the 15 judgments per person are 15 distinct items, then $\mathrm{TI}_{m_1}(aj) = \mathrm{Corr}_{i \in I}(\cdot)$ is computed over $|I| = 15$. A Spearman at $n = 15$ has a 95% CI of roughly ± 0.25 even near $\rho = 0.9$, and much wider in the middle of the range. **Both nested correlations are small-n**, and the paper currently reports neither CI. Report both.

> **CORRECTED by the repo audit (§1.8):** the 5×15 decomposition was wrong — it was 5 *splits* $\times$ 15 rows, one rater. The inner correlation actually runs over $n = 72$ aligned items, not 15 (Spearman 95% CI roughly ± 0.2 at mid-range values — workable). The *outer* correlation over judges remains the binding power constraint. CIs on both are still owed.

Problem 3 — the open question the page ends on

> **Q: OK I don't remember how the $\hat{y}$'s were selected.** But had issues with leaderboard `[?]` or something. So let's figure this out so I can do this clearly.

This is not a minor bookkeeping gap. **The selection rule for the annotated $\hat{y}$'s determines what TI_{m_1} means and whether it is estimable at all.** Specifically:

If the $\hat{y}$'s were selected by…	Then TI_{m_1} is…
Uniform random over (task × agent)	An unbiased estimate over the deployment distribution. Best case
Stratified by agent	Unbiased *per agent*; pooling requires reweighting
Stratified by judge score, or by bug level	**Biased** — conditioning on the judge's own output is selection on the dependent variable, and it inflates or deflates correlation in a direction that depends on the strata
Filtered to compiling / non-degenerate outputs	Restricted range. Systematically **attenuates** correlation
Convenience / whatever was on hand	Unknown estimand. Not reportable

And the **assignment design** matters just as much:

- **Crossed** (all 5 people judge the same 15 items) ⇒ 15 unique $\hat{y}$'s, inter-rater agreement computable, $\rho_{\max}$ estimable, persona stratification possible.
- **Nested** (each person judges a different 15) ⇒ 75 unique $\hat{y}$'s, **one rating each**, no inter-rater agreement, no $\rho_{\max}$, and TI_{m_1} against a "panel aggregate" is not well defined because there is no panel per item.

The $10 + 5$ arithmetic suggests it may be **partially** crossed, which is the worst case for analysis clarity because it requires a mixed-effects treatment rather than a plain correlation.

> **This must be answered from the repository, not from memory, before any resubmission.** (See also §1.7's 75-vs-150 conflict, which sharpens what Q10 must reconcile.) A repo-tracing agent prompt for exactly this question is provided as a companion artifact (`agent-prompt-yhat-selection.md`). Until it is answered, every number that depends on TI_{m_1} — including $\rho \approx 0.88$ — has an unspecified estimand.

1.7 TI_2' concretized: paired data, the three K's, and the size map (`39217`/`39218`)

Gold criterion, ratified form. $\mathrm{TI}_{2'}$ = correlates with human judgments **and** is stable, over all agents/solvers in the benchmark; PTI approximates it ("this is what we want to approximate with PTI"). Corrected boxed formula (three bugs in the handwritten version: missing orientation flip — the same $\Delta 4$ bug fixed once already on `37253`; STD subscript must be the judge's own scores since P_θ *is* the judge and stability is the label-free conjunct P_4; Corr must be clipped/mapped to $[0, 1]$ before any GM):

$$\mathrm{TI}_{2'}(aj) \;=\; \Big(\mathrm{clip}_{[0,1]}\big(\mathrm{Corr}(D_{\hat{S}_{aj}}, D_{\hat{S}^*_H})\big) \;\cdot\; \big(1 - \mathrm{STD}[D_{\hat{S}_{aj}}]\big)\Big)^{1/2}$$

Pairing rule (author's own margin: "has to be on the same examples" — correct). Corr is defined only over paired observations: the same candidate (i_b, i_a, i_{k_a}) scored by both sides. The $\forall x, y^*$ in the handwritten $\mathrm{TW}_{2'}$ is misplaced — items are what the correlation runs *across*, not quantified over (pointwise syntax on a correlational statement). The real residual choice is the $\forall a$: **pooled-across-agents Corr** (what the paper does) vs. **per-agent Corr, all required high** (strictly stronger; the "multiple plots" idea from `37252`). Pick one and state it. Also $aj = h_j \to aj \approx_\Pi h_j$.

"What is K" — three distinct K's, never pairwise aligned: K_a = agent generations per problem (paper: 3); K_{aj} = judge repeats (paper: $n_{\text{repeats}} = 3$, aggregated); K_{h_j} = human passes (paper: 2). Repeats never pair across sides; aggregate repeats first, then correlate at item level against each label view (pass1/pass2/avg) — exactly the paper's protocol.

Size map — idealized notes vs. actual paper data:

Notes symbol	Idealized	Paper actual
$\|b\|$	~800	857 theorems (HE 387, ES 278, CS 68, SS 112, RCS 12). Table rows CS/hE/ES/SS/RCS map 1-to-1; row "IS" matches no subset `[?]`
$\|L_A\|$	~10	3 generations/problem on the annotated slice; solver identities unenumerated → repo Q
$\hat{Y}$	~8,570 candidates	annotated slice: **75 rated pairs = 25 problems × 3 generations; 72 analyzed** (one problem's 3 rows drop on a stale filename — see §1.8; my earlier "+3 from duplicated legacy entries" guess was **wrong**, that's an unrelated 3)
$\|H\|$	named set	**not recorded**; only "two passes" (stricter/lenient); passes ↔ raters unknown
human judgments total	—	**75 × 2 = 150 ratings, 0–5 scale** — the entire human dataset behind every headline number
$\text{Corr}(D, D)$	—	per-judge validation human Spearman over ~72–75 rows; best single judge 0.480 (avg) / 0.559 (pass1); human–human inter-pass 0.249 [-0.015, 0.489] = the noise-ceiling anchor
bridge judges	—	8 configs (main), 13 (expanded, $\rho = 0.769$ avg labels)

[OK] Conflict RESOLVED by the repo audit (§1.8): it was **5 splits × 15 rows** (5 problems per split × 3 agents), one rater — not 5 people × 15. Total: **150 rated labels (75 pairs × 2 rubric rounds), 144 analyzed (72 × 2).**

[!] P_4-both-sides guard applies to $\text{TI}_{2'}$: putting STD inside the gold while P_4 sits in the proxy re-creates threat #21. Resolutions (either is fine, mixing is not): **(a)** bridge = $\text{PTI}_{\{1,2,3\}}$ vs TI_{m_1}, with $\text{TI}_{2'}$ reserved as the deployment/decision criterion — **recommended, matches §3.5(d)**; or **(b)** gold = $\text{TI}_{2'}$ with proxy strictly $\text{PTI}_{\{1,2,3\}}$.

Paper-side aggregator reference (the orientation-fixed, clipped forms the notes should inherit): $C_f = \frac{1}{|\mathcal{F}|}\sum_f \text{clip}(\frac{1-\rho_f}{2}, 0, 1)$; $\text{GM} = \text{mono}^{0.4} \cdot \text{stab}^{0.6}$; $\text{stab} = 1 - \text{flip_rate}$; $\text{TI}_{\text{var}} = \exp\left(\frac{1}{3}\log C_f + \frac{1}{3}\log(1 - \widetilde{\text{VarCorr}}) + \frac{1}{3}\log(1 - \widetilde{\text{VarScore}})\right)$. Note the paper's TI_{var} contains **no human term** — it is the proxy (PTI) side; the combined gold $\text{TI}_{2'}$ is a **new object** the paper never computes. The paper's gold is TI_{m_1} (per-judge human Spearman) alone.

1.8 Effective parameters, $b' \subseteq b$, and the audited ground truth

1.8.1 The effective-parameter split (`39296/39297`)

The pages separate what each object *actually* depends on — and expose the asymmetry that IS the amortization argument:

$$\mathrm{PTI}(\underbrace{b}_{\text{full VB possible}}, L_A, aj, P) \overset{\text{s.t. } \approx \text{ (e.g. correlates)}}{\rightsquigarrow} \mathrm{TI}_{2'}(\underbrace{b'}_{\subseteq b}, L_A, aj, H)$$

- **TI's effective params are** (b', L_A, aj, H) plus repeat counts (k_a, k_{aj}, k_H) — and the $|b|$ in earlier tensor notation **overstated coverage**: "my notation with $|b|$ assumes we got agent/solver generations & their AI-judge scores for all of b — but really we only did $b' \subseteq b$ … duh, cuz human evaluations are expensive. That's the whole point of PTI." Every D-tensor on the gold side is $[|b'| \times |L_A| \times k_a \times (\cdot)]$.
- **PTI's effective params are** (b, L_A, aj, P) — "full VB possible!", because once P is specified the property pipeline is automatic. (The page's own TODO — "don't rly know yet if that's a good idea" — resolves cleanly: computing PTI on full b is fine and is the point; the caveat is that its *validation anchor* lives on b'.)
- **This names a fourth generalization axis:** the bridge is validated on b', but PTI is *used* on $b \setminus b'$ — a **within-benchmark item-level extrapolation** ($b' \to b$), alongside the judge, agent, and benchmark axes. State it in the paper; do not let it ride silently.
- **The 1 vs $|H|$ slot:** the judge tensor carries 1 where the human tensor carries $|H|$ — one entity is being validated against a pool. TW with $b' \subseteq b$ boxed as "care!": the criterion is only ever *measured* on the subset.
- k_x (a prompt dimension) flagged and deliberately deferred: "might be prompt, but it's distracting me right now."
- **Lay naming for P_1:** "reflexivity is too mathematical, hard to explain to a lay person (a VC, or Rylan in the car)." Candidate on the page: "equality?". Recommendation: use **identity** — it is already the paper's word (Fig. 13b, "identity variant breakdown"), with "self-consistency" as the spoken gloss.

1.8.2 The audited ground truth (Claude Code forensics, `cert-judge@main`, 13 Aug 2026)

The repo audit answers the forensics prompt. Findings, condensed (full write-up: `experiments/26_human_label_dataset_audit/READ` single source of truth: `results/aligned_panel_n75.tsv`):

Question	**Audited answer**
b'	**25 problems × 3 generations = 75 rated pairs; 72 analyzed** (24 problems). The paper's "24 × 3 = 75" never multiplies out
Where the 3 go	They **drop**, not add: `0_unsafeMultiply_safe` was renumbered to `1_…` in the dataset; the label rows kept the old name → `missing_gold` × 3 (`skipped_pairs.csv`; `summary.json`: `n_union_rows = 72`)
Fig. 11's three 1.5s	An **unrelated** 3: `4_json_decoder` duplicated in the lenient file (scored 1 and 2 → avg 1.5), collapsing back to the same 75. Two different 3s
$\|H\|$ on the panel	**1 (Ethan).** Two *rubric-strictness rounds* by one rater — not two raters. No rater column exists in any of the 16 label TSVs (identity was filename-only, then anonymized); pooled files cannot be split by rater
pass1/pass2	**Inverted vs. chronology** (sha-verified: paper pass1 ≡ the annotator's *second* pass). Read as strict/lenient, never first/second

Question	**Audited answer**
Human labels total	**150 rated (75 × 2), 144 analyzed (72 × 2)**
"5 × 15" (`38707`)	**5 splits × 15 rows** (5 problems/split × 3 agents), one rater — resolved
$\hat{y}$ selection	**Stratified 5-per-split, hand-picked** (e.g. cs_set = exactly the 5 classic sort/search problems of 11 on disk), **frozen 2026-04-07** (commit `3a7b75c`), **no script, no seed**, not compile-filtered
Agent identities on the slice	`AGENT_MAP = {0: codex, 1: claude, 2: llama}` — the `39218` table's Leanstral/Qwen were *judges*, not the annotated agents
Full candidate pool	471 files = **157 problems × 3** (all of VB v1 at *problem* granularity; the 857 figure is *theorems*) — the panel is 25 of 157
The 5-rater dataset	A **separate population**: 343 labels over 78 pairs, 5 raters, *revised* rubric; correlates with the calibration panel at only $\rho \approx 0.13$ — **documented in-repo as the cause of the exp43 held-out null**
Seeded-selection template	Exists but never run: study 41, `selection.json` seed 41, $n = 30$; 0% collected. The reviewer-proof path if a principled sample is demanded
Number hygiene	`main.tex` MAE/MSE (0.778/1.139) uses first-of-duplicates; cached mean-of-duplicates gives 0.7708/1.1215 — one convention must be chosen and regenerated. G.3's "~72–75" hedge is determinate: **72**

Corrections to these notes, on the record: (i) my "+3 = duplicated legacy entries" reading of the 72–75 wobble was wrong (two different 3s); (ii) the "inner correlation at $n = 15$" warning rested on the wrong 5×15 decomposition — inner n is **72**; (iii) "power is probably the explanation for the held-out null" is demoted — **criterion drift ($\rho \approx 0.13$ across pools) is the documented cause**, power remains binding on top; (iv) $\rho_{\max}$ (within-persona inter-person ceiling) is **not computable on the calibration panel** ($|H| = 1$) — the 0.249 inter-pass figure is one person's *rubric-stability*, an even harsher ceiling statement: the criterion moves by $1 - 0.249$ worth of rank noise under a strictness change alone. Inter-person ceilings must come from the 343-label pool; (v) crossed-vs-nested resolves trivially (fully crossed: $1 \times 75 \times 2$); (vi) the "union destroys the rater index" worry *understated* it — no per-row rater field ever existed; add one to the schema going forward.

Interpretive consequence worth a paragraph in the paper: with $|H| = 1$, the panel's TI_{m_1} is *one person's criterion under a stated rubric* — which, via the $h_j = (\text{person}, \pi)$ decomposition, is a perfectly legitimate scoped criterion, but it must be *said*, and the rubric is part of the criterion's identity (the 0.249 and 0.13 numbers are both measurements of exactly that).

1.8.3 Decision forks (author calls)

1. **72 vs 75 in the paper.** Option A: reword — "75 rated pairs from 25 problems × 3 generations; 72 (24 problems) enter all analyses because one problem's gold artifact is indexed under a pre-renumbering filename." Option B: rename the 3 stale label rows to `1_unsafeMultiply_safe` and regenerate EXP16/EXP18, making $n = 75$ literally true. **Recommendation: B**, because threat #10 already commits to regenerating all headline numbers from a single artifact run before resubmission — the marginal cost is near zero and it deletes the discrepancy instead of footnoting it. If no regen were planned: A.
2. **Rename pass1/pass2 → strict/lenient** throughout paper and code. Removes the chronology inversion trap permanently.

3. **Commit the audit folder** (`experiments/26_human_label_dataset_audit/` is currently local-only) so every pointer above resolves for the team — approve Claude Code's push.
4. **Before editing `main.tex`, re-run the audit scripts** (`audit_human_labels.py`, `reconcile_paper_numbers.py` — pure stdlib): the findings above are agent-reported and the standing verification discipline applies to friendly agents too.

Part II — Why Definition 1 cannot be checked directly

2.1 The cost object

> **Q: What trust index makes sense to design?** ↪ i.e. *what is expensive* in the "certification" of AI judges? ↪ **Human labels.**

Checking Definition 1 directly for one judge requires, for every benchmark b, every agent a, and every item (x, y^*), a matched human judge h_j to score $a(x)$ — repeatedly, if M involves a variance term. Cost is

$$C_{\text{direct}} \;=\; |B| \cdot |A| \cdot n_b \cdot |H| \cdot R \cdot c_{\text{human}}$$

where R is repeats per item and c_{human} is the per-rating human cost. Every factor except c_{human} grows. This is the **time-complexity argument** the TODOs sheet asks for:

> "we need to explain the problem we are solving precisely (I think it is **bench-scalable-oversight**); with **time complexity arguments**."

Name the problem: benchmark-scalable oversight. The claim to defend is that certification cost must not grow with $|A|$ or $|AJ|$.

Refined by §1.6(b): labels are not eliminated, they are de-indexed. The honest cost statement is

$$C_{\text{amortized}} \;=\; \underbrace{|\Pi| \cdot n \cdot c_{\text{human}}}_{\text{persona conditioning, once per persona}} \;+\; \underbrace{|AJ_{\text{val}}| \cdot |I| \cdot c_{\text{human}}}_{\text{bridge validation, once}} \;+\; \underbrace{|AJ| \cdot |A| \cdot |Q| \cdot c_{\text{API}}}_{\text{certification, per judge and agent}}$$

with $c_{\text{API}} \ll c_{\text{human}}$. The first two terms are **constant in the number of agents and in the number of judges subsequently screened**; only the third term grows, and it grows in API calls. Say it this way. Do **not** say "no human labels" — §1.6 shows that is false, and it is the kind of claim a reviewer disproves from your own method section.

2.2 The objection you wrote against yourself

> ⇒ **"but then, why don't you just use the human labels you had?"**

This is the sharpest line in the entire stack and it is the first question a reviewer will ask. It has a structural answer, and it should appear in the introduction in these terms:

1. **Old labels are indexed to the (a, b) pairs that existed when they were collected.** A new agent a' produces artifacts off the support of the labelled set — new failure modes, new styles, new ways of being wrong. The old labels score old outputs; they say nothing about a''s outputs.
2. **A new judge aj' has never been validated.** Re-validating it against old labels *is* the per-judge annotation loop the framework exists to remove — and it is only sound if the old labelled items remain representative, which is precisely what a new agent breaks.
3. TI **is label-free by construction.** Every property in Part III is measured by perturbing an artifact and re-querying the judge. Recomputable for any new (aj', a') at API cost.

4. **Therefore: human evals are done once.** The fixed annotation cost is amortized across the judge population and the agent population. Cost becomes $C_{\text{amortized}} = C_{\text{one-time bridge}} + |AJ| \cdot |A| \cdot c_{\text{API}}$, with $c_{\text{API}} \lll c_{\text{human}}$.

That is the thesis. State it as the answer to this question, do not leave it implicit.

Part III — The certificate

3.1 Design principle

The certificate must be computable **without** P_H. So it cannot measure Definition 1. It measures **necessary conditions** that any judge satisfying Definition 1 must also satisfy — behavioural invariances that follow from the judge being a *coherent scoring function at all*.

Definition 3 (behavioural property). A property $P_k : AJ \to [0, 1]$ is admissible iff

- **(a) Label-free:** $P_k(aj)$ is computable from queries to aj alone, given a probe suite.
- **(b) A priori necessary:** it is derivable from a stated metamorphic taxonomy *before* seeing any human data — not selected because it correlates with human agreement.
- **(c) Oriented:** higher is better, $P_k \in [0, 1]$, with $P_k = 1$ meaning "no violation."

Condition (b) is the fix for the **selection-circularity** objection raised against the current draft. If properties are curated by their correlation with human agreement and that same correlation is then the headline result, the argument is circular. Deriving them from a taxonomy and **locking the set before touching the held-out reviewer pool** breaks the loop.

3.2 The four properties

#	Property	Metamorphic relation	Violation looks like
P_1	**Identity / reflexivity**	T semantics-preserving $\Rightarrow s(T(\hat{y})) = s(\hat{y})$	Score moves under rename / reorder / restyle / α-renaming
P_2	**Bug monotonicity**	$\hat{y} \succ \text{bug}_1(\hat{y}) \succ \cdots \succ \text{bug}_K(\hat{y}) \Rightarrow$ scores non-increasing	Local reversal on the bug ladder
P_3	**Specification monotonicity**	Removing obligations/tests/theorems weakens the artifact $\Rightarrow$ score non-increasing	Judge rewards a weaker spec
P_4	**Stability**	Fixed input, repeated queries / paraphrased prompt / temperature jitter $\Rightarrow$ same score	High std across repeats

P_1 and P_2 are exactly what Figures 13 and 14 of the current draft measure:

- **Fig. 13(a)** — main low-variance monotonicity curves (core GPT-5.3 judges, robust repeat aggregation), judge score vs. bug level L_0 (clean) $\to L_4$ (most degraded); series `P8_STRICT_B`, `P4_CHECKLIST`, `P1_BASELINE`. $\to P_2$**.**
- **Fig. 13(b)** — failure-mode isolation: identity-variant breakdown with `ALPHA` shown separately (`RENAME`, `REORDER`, `STYLE`, `ALPHA`). $\to P_1$**.**
- **Fig. 14** — local bug-ladder reversal rates by judge; surfaces where monotonicity fails locally. $\to P_2$ **violation diagnostic.**

3.3 The Trust Index

Definition 4 (Trust Index).

$$\boxed{\mathrm{PTI}_p(aj) \;=\; \mathrm{GM}(P_1(aj),\, P_2(aj),\, P_3(aj),\, P_4(aj)) \;=\; \Big(\prod_{k=1}^{4} P_k(aj)\Big)^{1/4} \;\in\; [0,1]}$$

Why geometric mean: SOFT-AND semantics — corrected. Earlier drafts of these notes said a failing property "collapses" the GM. That is only true at exactly zero: $\mathrm{GM}(1,1,1,0) = 0$, but $\mathrm{GM}(1,1,1,0.5) \approx 0.84$ — a judge can fail one property *substantially* and keep a high-looking aggregate. (Credit: the GPT review caught this; the arithmetic is not arguable.) The precise statement:

- GM is **zero-annihilating** and penalizes imbalance more than the arithmetic mean — a *soft* AND, and still the right choice over AM for exactly the reason above ($P_2 = 0$ must kill the score, and under AM it does not).
- **Hard gating is a separate mechanism**: the screening decision uses per-property thresholds on confidence bounds — FAIL-SCREEN when $\mathrm{UCB}(\hat{P}_k) < \tau_k$ for any required k — not the aggregate.
- **The auditable object is the property vector $\mathbf{P}(aj) = (P_1, \ldots, P_4)$**, always reported in full; PTI_p is its compact ranking summary, nothing more. If worst-case conformance matters, report $\min_k P_k$ alongside.

The handwritten two-argument form was

$$\mathrm{GM}\Big(\underbrace{\mathrm{Corr}(\hat{s}, s^*)}_{\text{high}},\; \underbrace{1 - \mathrm{std}(\hat{s}, s^*)}_{\text{low std}\Rightarrow\text{high}}\Big) = \text{high}$$

with the first draft writing Var in the second slot and then correcting it to the orientation-fixed $1 - \mathrm{std}$ (arrow labelled $\Delta 4$ / P4). Note that the **first slot of that expression is not label-free** — it contains s^*. Definition 4 replaces it with the four label-free properties; the correlation term moves to the *right-hand side of the bridge* in Part V, where it belongs. This is a real correction, not a cosmetic one: **the Trust Index must not contain human scores, or the whole cost argument collapses.**

3.4 What the screen does and does not license — CORRECTED

Retraction first. Earlier versions of these notes claimed "sound rejection by construction": low $\mathrm{PTI}_p \Rightarrow$ not trustworthy, as a proof. **That claim is false at the ratified target** m_2, and the GPT review is right to kill it. Under $\mathrm{TW}_2 = [\mathrm{Corr}(S_{aj}, S_{h_j})\ \text{high}] \wedge [\mathrm{Var}\ \text{low}]$, correlation is an *aggregate* statistic: a judge can carry high human correlation while exhibiting a few local bug-ladder reversals, one rename sensitivity, or a spec-ladder violation on an odd item. So P_k low $\not\Rightarrow \mathrm{TI}_{m_1}$ low *as a matter of logic*. Necessity of P_1–P_3 holds only for **pointwise** trustworthiness, not correlational.

The m-ladder stratifies which claims are logical and which are empirical:

Target	Status of P_1–P_3	Rejection semantics
Pointwise: $\lvert J(z) - H(z)\rvert \le \varepsilon\ \forall z$ (strong, m_0-flavoured)	**Necessary under assumptions** — see the falsification lemma below	Theorem-backed, conditional
m_2 (correlational + stable — **the ratified target**)	**Not necessary; empirical predictors**	Definitional (Route A) or evidential only

Route A (definitional). Fold the properties into the target: $\mathrm{TW}(aj) := [\mathrm{TI}_{m_1} \ge \tau_A] \wedge \bigwedge_k [P_k \ge \tau_k]$. Then a probe violation rejects **by definition** — at the price that "properties predict alignment" becomes a separate empirical question, which must still be reported.

Route B (conditional falsification lemma — the theorem opportunity). Suppose (i) pointwise ε-alignment: $|J(z) - H(z)| \leq \varepsilon$ on the probe domain, and (ii) a **human margin** on the transformation: $H(x) - H(Tx) \geq \delta$ (humans really do score the bugged artifact lower by at least δ). Then

$$J(x) - J(Tx) \;\geq\; \delta - 2\varepsilon,$$

so an observed reversal larger than $2\varepsilon - \delta$ **refutes** ε-alignment; likewise if T is human-invariant, $|J(x) - J(Tx)| \leq 2\varepsilon$. One-line proofs, but honest ones. The paper sentence becomes: *"under stated metamorphic-validity and margin assumptions, sufficiently large probe violations falsify pointwise ε-alignment"* — never "sound rejection by construction."

> **Concrete new experiment this unlocks (cheap):** the margins δ are currently *assumed* — per the workshop paper's own Table 1, humans never rated the perturbed ladder variants. Collect a small human sample **on ladder items** (a few dozen ratings) to estimate δ per rung and per transform. That single collection turns Route B from an assumption into a grounded theorem with measured constants, and it also directly tests whether ALPHA renaming is truly human-invariant (the standing P4 failure mode becomes interpretable).

- **Acceptance is unchanged**: high PTI_p never establishes trustworthiness at any m; it is licensed as evidence only via the bridge (Part V), within the validated population.

So the instrument is, by construction, a **screen / falsification test**, and only the empirical bridge in Part V can upgrade it to a predictive ranking instrument — and only within the population it was validated on. The ratification page states this in one line: *"if it's high ⇒ 'certifies' the ai judge — cuz we demonstrated that PTI **does** correlate with TI. key!"* — acceptance in scare quotes, purchased by the bridge, never free. Write it this way. The desk rejection came from an abstract that claimed the strong reading while the body reported the screen finding no qualifying signal; stating the screen semantics up front makes that combination coherent instead of contradictory.

3.5 One conjunct of TW_2 is directly estimable — and this shrinks what the bridge must carry

Writing the rename out (§0.4) makes something visible that was hidden while both objects were called "trust": TW_2 **is a conjunction, and one of its two conjuncts is already label-free.**

$$\text{TW}_2(aj) \;\equiv\; \underbrace{[\ \text{Corr}(S_{aj}, S_{h_j})\ \text{high}\ ]}_{\textbf{needs humans} = \text{TI}_{m_1}(aj)} \;\wedge\; \underbrace{[\ \text{Var}_{P_\theta}[s]\ \text{low}\ ]}_{\textbf{label-free} = P_4}$$

The second conjunct is not a *proxy* for trustworthiness under TW_2. It **is** one conjunct of TW_2, and P_4 **estimates** it directly with zero annotation (finite repeats estimate the population variance; they do not verify it exactly — report the estimate with its CI). Wording discipline: say "one conjunct is directly estimable label-free," never "half is free" — conjuncts carry no equal quantitative weight merely by being two.

Three consequences, all favourable:

(a) The empirical claim shrinks. The bridge only has to carry $P_1, P_2, P_3 \to \text{TI}_{m_1}$. That is a strictly smaller claim than "PTI_p predicts trustworthiness," and given that the pre-registered bridge screen did not clear its bar, making the claim smaller is the difference between a paper and no paper. Report it as: *one conjunct of the definition is directly verified; the other gets a screening instrument whose predictive validity we measure and report honestly.*

(b) No circularity in the bridge — state this explicitly. P_4 appears in PTI_p but **not** in TI_{m_1} (which is the correlation conjunct only). So correlating PTI_p against TI_{m_1} is clean. But it would **not** be clean to correlate PTI_p against a composite "TW_2 score" that itself includes the variance term — that puts P_4 on both sides. Whoever implements Algorithm 5 must be told this in a comment, because it is exactly the kind of thing that gets silently introduced during a refactor.

(c) A cheap robustness result you already have the data for. Report the bridge twice: once with $\mathrm{PTI}_p = \mathrm{GM}(P_1, P_2, P_3, P_4)$ and once with $\mathrm{PTI}_{p\setminus\{4\}} = \mathrm{GM}(P_1, P_2, P_3)$. If the bridge is materially stronger without P_4, that is evidence P_4 is doing definitional work rather than predictive work — which is what §3.5 predicts, and it would be a nice confirmatory detail rather than an embarrassment.

3.6 Two jobs, two validations: screen vs. surrogate (adopted from the GPT review)

PTI_p has been doing two jobs with one validation story. Separate them; each gets its own metrics, and the paper can legitimately study both.

	Screen (gate before humans)	**Surrogate** (predict the human criterion)
Claim	catches unacceptable judges cheaply	$\widehat{\mathrm{TI}}_{m_1}(aj) \approx \mathrm{TI}_{m_1}(aj)$ numerically
Decision	FAIL-SCREEN / PASS-SCREEN / **DEFER**	a predicted score, with interval
Validation metrics	false-pass rate, false-reject rate, defer rate, coverage, **human ratings saved at a fixed false-pass tolerance**	held-out rank correlation, MAE / calibration error, prediction intervals, transfer across families / prompts / personas / agents
Can succeed while the other fails?	yes — a screen is useful even if it cannot rank the acceptable	yes — a surrogate can rank without any sound rejection rule

Two adoption notes:

1. **Decision utility is the missing headline metric.** "Scalable oversight" becomes an operational number: *at false-pass tolerance α, the screen avoids X% of human validations.* Report that, plus sensitivity to shift and a re-validation horizon. A correlation plot alone does not answer the question the paper poses.
2. **A calibrated surrogate map $f_{\mathrm{cal}} : \mathbf{P} \mapsto \widehat{\mathrm{TI}}_{m_1}$ is the right eventual architecture — and premature at $n_{\mathbf{judges}} \approx 8$.** Fitting even isotonic regression over four features on eight judges is overfit theatre. At current scale, the honest object is the pre-specified aggregate (PTI_p) with its raw held-out correlation; graduate to f_{cal} with hat-notation semantics when $|AJ|$ reaches ~40. (This also resolves the earlier hat-notation dispute: a hat is *earned* by a calibration map; the raw aggregate never gets one.)

Inconclusive ≠ negative. One more adopted point: a wide CI spanning strong-positive through negative is an *inconclusive* result, not a negative one. For the negative reading to be a finding, pre-register a **smallest effect size of interest** (SESOI, e.g. $\rho_0 = 0.5$: below this the screen saves too little to deploy) and run an equivalence / non-inferiority test against it. That converts "we failed to detect" into "we bound the effect below the deployable threshold" — which *is* publishable.

Part IV — The screening & validation protocol

Protocol name (adopted): amortized judge validation. "Certification" is retired from the scientific claim — both because TI_m is itself the better certification when affordable (the author's own objection), and because the standards meaning of certification (ISO conformity-assessment usage: written third-party assurance of meeting specified requirements) is simply not what this protocol does; it screens, measures, and validates a surrogate. "Amortized judge validation" is also *continuous with the workshop paper's own contribution #1* ("an amortized trust-validation protocol"), so the rename reads as sharpening, not pivoting. `cert-judge` stays as the repo slug for history.

All algorithms below are label-free except Algorithm 5, which is run **once**.

Algorithm 1 — Probe suite construction

```
INPUT : benchmark b, agent set A, ladder depths K_bug, K_spec, repeat count R
OUTPUT: probe suite Q

Q ← ∅
for each agent a in A:
    for each item (x, y*) in b:
        ŷ ← a(x)                                  # the artifact under judgment
        # --- P1: identity / semantics-preserving variants ---
        for T in {RENAME, REORDER, STYLE, ALPHA}:
            Q ← Q ∪ {(x, y*, T(ŷ), tag=P1, variant=T)}
        # --- P2: bug ladder, monotone degradation ---
        ŷ_0 ← ŷ
        for k = 1..K_bug:
            ŷ_k ← inject_bug(ŷ_{k-1})             # strictly more degraded
            Q ← Q ∪ {(x, y*, ŷ_k, tag=P2, rung=k)}
        # --- P3: specification ladder, obligation removal ---
        sigma_0 ← spec(x)
        for k = 1..K_spec:
            sigma_k ← drop_obligation(sigma_{k-1})        # strictly weaker spec
            Q ← Q ∪ {(x, y*, ŷ under sigma_k, tag=P3, rung=k)}
        # --- P4: stability replicates ---
        Q ← Q ∪ {(x, y*, ŷ, tag=P4, repeat=1..R, prompt_paraphrase, temperature jitter)}
return Q
```

Pre-registration requirement. The ladders, the transform set, K_{bug}, K_{spec}, R, the tolerance τ_{id}, and the aggregation rule are all fixed and committed **before** any judge is run. Otherwise property selection re-enters through the back door.

Algorithm 2 — Property estimation

```
INPUT : judge aj, probe suite Q
OUTPUT: P1, P2, P3, P4 in [0,1]

# ---- P1: identity / reflexivity ----------------------------------------
D_id ← { | s_aj(x,y*,T(ŷ)) - s_aj(x,y*,ŷ) | : (.,tag=P1) in Q }
P1 ← 1 - mean(D_id) / range(S)                    # or: frac(D_id ≤ τ_id)

# ---- P2: bug monotonicity ----------------------------------------------
# ladder order is known by construction: rung 0 ≻ 1 ≻ … ≻ K
τ_b ← mean over ladders of Kendall-τ_b( median score per rung, -rung index )
P2 ← (τ_b + 1) / 2                                 # map [-1,1] → [0,1]
# diagnostic (Fig. 14): local_reversal_rate ← frac(ladders with ≥1 adjacent increase)

# ---- P3: specification monotonicity ------------------------------------
# identical estimator on the obligation-removal ladder
P3 ← (Kendall-τ_b(spec ladder) + 1) / 2

# ---- P4: stability -----------------------------------------------------
sigmã ← mean over probe points of std({ s_aj repeat 1..R })
P4 ← 1 - sigmã / range(S)

return P1, P2, P3, P4
```

Notes on the estimators:

- **Everything is normalized to** $[0, 1]$ **before aggregation** — GM is only meaningful on a common scale, and a negative entry makes the product meaningless.
- P_2, P_3 **use rank correlation against a known order**, not a regression slope. The ladder order is a design fact, not an estimate, so a rank statistic is the right tool and it is robust to the judge's arbitrary score scale.

- **Report the local reversal rate alongside** P_2 (Fig. 14). A judge can have a good global τ_b and still reverse locally, and local reversal is the failure that matters operationally.

Algorithm 3 — SCREEN (label-free, per judge; formerly CERTIFY)

```
INPUT : judge aj, benchmark b, agent set A, threshold τ
OUTPUT: TI in [0,1], decision in {REJECT, PASS-SCREEN}, per-property report

Q               ← BuildProbeSuite(b, A, …)          # Algorithm 1, cached across judges
P1,P2,P3,P4     ← EstimateProperties(aj, Q)         # Algorithm 2
TI              ← (P1 . P2 . P3 . P4)^(1/4)
CI              ← bootstrap over items and agents   # resample probe points, recompute TI
decision        ← REJECT if upper(CI) < τ else PASS-SCREEN
return TI, CI, decision, {P1,P2,P3,P4}
```

The probe suite is cached and reused across judges. That is what makes the marginal cost of certifying a *new* judge equal to $|Q|$ API calls and nothing else. It is the operational core of the amortization argument.

Always report the per-property vector, not just TI**.** The GM is lossy by design; the diagnostic value is in which property failed.

Algorithm 4 — Certifying a *new agent* (the $\forall a$ axis)

```
INPUT : certified judge aj, new agent a'
OUTPUT: whether aj's screen result extends to a'

Q'  ← BuildProbeSuite(b, {a'}, …)                   # probes built on a''s own artifacts
P'  ← EstimateProperties(aj, Q')
TI' ← GM(P')
return PTI' , and: screen result extends iff PTI' ≥ τ and |PTI' - PTI| within pre-registered slack
```

This is the step that answers *"why not just use the human labels you had?"* operationally: the probes are rebuilt **on the new agent's own outputs**, so the certificate is re-earned on the new distribution without a single new human rating.

Algorithm 5 — Bridge validation (label-requiring, run ONCE)

```
INPUT : judge populations AJ_train ⊎ AJ_test (disjoint), human panel H, benchmark b
OUTPUT: bridge estimate rhô with CI, and a decision on whether TI is predictive

# ---- one-time human collection -----------------------------------------
for each item i in a fixed sample I ⊆ b, for each agent a in a fixed sample:
    collect human ratings { s_h(i) : h in H }, persona-labelled

# ---- inner correlation: per judge, over ITEMS ----------------------------
for each aj in AJ_train ∪ AJ_test:
    Align(aj) ← Corr_{i in I}( s_aj(i) , aggregate_{h ≈_prod aj}( s_h(i) ) )

# ---- tuning on TRAIN only ------------------------------------------------
fix estimator details, τ, normalizations using AJ_train ONLY

# ---- outer correlation: over JUDGES --------------------------------------
rhô ← Corr_{aj in AJ_test}( TI(aj) , Align(aj) )
CI ← bootstrap over judges          # ← n = |AJ_test|, NOT |I|
return rhô, CI
```

> **[!] Power warning, and this is probably the explanation for the failed pre-registered screen.** The outer correlation's sample size is **the number of judges**, not the number of items. With $|AJ_{\text{test}}| \approx 8$–$12$, a Spearman ρ has a 95% CI roughly ± 0.5 wide — wide enough that a true ρ of 0.6 routinely fails a significance screen. **The fix is more judges, not more items.** Judges are prompt/model/persona configurations; they cost API calls, not annotation. Scaling $|AJ_{\text{test}}|$ from ~10 to ~40 is cheap and is the single highest-leverage change available before resubmission.

Adding items increases the precision of $\mathrm{TI}_{m_1}(aj)$ but does nothing for the power of the claim you actually want to make.

UPDATE (repo audit, §1.8): power is real but is no longer the leading explanation for the held-out null. The audit found the held-out human criterion came from a *different rater pool under a revised rubric* correlating at only $\rho \approx 0.13$ with the calibration panel — documented in-repo as the cause of the exp43 null. A bridge calibrated against criterion A and tested against criterion B with $\mathrm{Corr}(A, B) \approx 0.13$ fails by construction, at any judge count. The resubmission story is **criterion drift first, power second**: fix the criterion (one rubric, one pool, per claim), *then* the ~40-judge power fix applies.

[!] Validation vs. test. The $\rho \approx 0.87$–0.88 figure currently in circulation is a **validation-set** number — the same human ratings were used to tune and to report. Label it "validation Spearman" everywhere it appears and put the held-out protocol in an appendix as future work. This is a known integrity item; it is not negotiable.

Part V — The bridge, and what the paper must actually prove

Three distinct claims, which the current draft conflates:

Claim A (definitional). aj is (M_4, ε)-trustworthy. *Requires humans. Not directly checkable at scale. This is the thing we want.*

Claim B (certificate). $\mathrm{PTI}_p(aj) \geq \tau$. *Label-free, cheap, checkable for any judge and any agent. This is the thing we can compute.*

Claim C (bridge). PTI_p predicts TI_{m_1} across judges:

$$\mathrm{PTI}_p(aj) \;\approx_C\; \mathrm{TI}_{m_1}(aj) \qquad \text{over } aj \in AJ_{\text{test}},\quad b = \mathrm{VB}$$

This is the only empirical claim the paper has to establish, and it is established once.

The logical shape — **corrected** (the $A \Rightarrow B$ arrow of earlier drafts holds only for pointwise targets under Route B assumptions, *not* at the ratified m_2):

$$\underbrace{\text{Claim B}}_{\text{compute, cheap}} \xrightarrow{\text{Claim C (validated once)}} \underbrace{\text{evidence about Claim A}}_{\text{want}} \qquad \text{plus} \qquad \text{theorem-backed falsification under Route B assu}$$

At m_2, failing B is *evidence against* A (strength set by the bridge), not a refutation; refutation is available only under Route A (definitional) or Route B (pointwise + margins). Passing B never establishes A; only Claim C, plus the population it was validated on, licenses using B as evidence for A. **Say this explicitly in the abstract.** An abstract that says "TI strongly predicts human alignment" while §5.5 reports a pre-registered screen finding no qualifying signal is a self-contradiction visible in under a minute, and that is what produced the same-day desk rejection. The honest version — *"we define a label-free certificate, show it is a sound screen, and report that its predictive bridge did not clear our pre-registered bar under the judge populations tested, with a power analysis showing why"* — is a publishable measurement-lessons paper. The overclaiming version is not publishable at all.

5.1 Two papers

The handwritten split resolves the circularity cleanly:

> **1)** finding the "good judge" / **certifiable judge 2)** finding a Trust Index for (hopefully 1's) judge — a **Trust Index that removes the need of human labels**

Paper 1 fixes the judge, the property taxonomy, and the probe suite. Paper 2 validates Claim C on a **held-out judge population**, with the property set locked between them. Locking between papers is what makes the second paper's headline number non-circular — which is exactly the worry recorded on the Trust Index sheet:

> "this seems an optimization over the judges. That's fine with me. But what we need is — at least per cert-judge — a Trust Index that **works**. So human evals we do **once**. Generalization across $a \in A$ … is really important."

Optimizing over *judges* is legitimate. Optimizing over *properties* and then reporting the optimized fit as evidence is not.

Part VI — Threats to validity

#	Threat	Status	Mitigation
1	**Selection circularity** — properties partly curated by correlation with human agreement, then that correlation reported as the result	Open. Flagged by Sanmi	Derive $P_1..P_4$ from a stated metamorphic taxonomy as a priori necessary conditions; lock before held-out reporting; two-paper split
2	M_2 **vs.** M_4 **conflict**	Unresolved in the notes	Choose M_4. Never claim "matches human variance" and "is low variance" in the same document
3	**Persona precondition ignored in practice**	Likely cause of observed sign flips	Persona-label the human panel; compute TI_{m_1} only within persona-matched pairs; report diverse-panel and narrow-panel results separately with the precondition stated
4	**Outer-correlation power**	Almost certainly binding	Increase $\lvert AJ_{\text{test}}\rvert$ to ~40+; report the power analysis; bootstrap over judges
5	**Validation reported as test** ($\rho \approx 0.88$)	Known integrity item	Relabel as validation Spearman; pre-registered held-out protocol in appendix
6	**TI / human index mismatch**	Notational, but fatal if published	Two nested correlations, explicitly typed (Part 0.3)
7	**Necessary** $\neq$ **sufficient**	Currently overclaimed	State screen semantics: sound rejection, no guaranteed acceptance

#	Threat	Status	Mitigation
8	**Sampling → expectation relaxation**	Acknowledged in the notes	*"to remove the sampling; otherwise we'd need a guarantee in probability."* State it as a relaxation, do not smuggle it
9	**Scope creep to all benchmarks**	Author's own answer is "just VB"	Claim VeriBench only
10	**Number inconsistencies paper vs. artifact** (Holm p, LOO p)	Known	Regenerate all numbers from a single artifact run before resubmission
11	**"No human labels" overclaim**	Would be false as stated	§1.6(b): labels enter at persona conditioning *and* bridge validation. Claim de-indexing ($O(1)$ in agents and screened judges), not elimination
12	**Persona-conditioning overfit at** $n \approx 10$	Unquantified	Hold out part of D_{h_j}; report method performance as a function of n; if $\mathcal{M}$ does prompt search on D_{h_j}, that is a second circularity site and needs its own lock
13	T_2 **/** T_3 **target confusion**	Probable cause of sign flips	Persona-label existing raters; recompute TI_{m_1} within strata; see whether the flip resolves
14	**No noise ceiling reported**	Missing	Estimate $\rho_{\max}$ within persona; report $\mathrm{TI}_{m_1}/\rho_{\max}$
15	**Definition over** A**, experiment over** L_A	Live gap	Parameterize Definition 1 by the agent set; claim the L_A version everywhere including the abstract
16	$\hat{y}$**-selection rule unknown** (`38707`)	**Blocking.** Author cannot recall it	Trace it from the repo before resubmission (companion agent prompt). Until known, TI_{m_1} has no specified estimand
17	**Inner correlation also small-**n ($\lvert I\rvert \approx 15$)	Newly visible	Report CIs on $\mathrm{TI}_{m_1}(aj)$, not just the point estimate
18	D_H^S **written as a union**	Destroys the person index	Store as an indexed family $(h_j, x, \hat{y}, s)$; verify against what the code writes

#	Threat	Status	Mitigation
19	**Possible partially-crossed design** (10+5 batches)	Unknown	If partially crossed, a plain correlation is the wrong estimator; use mixed effects
20	**"Trust Index" name asserts the conclusion**	Being fixed	Rename to Proxy Trust Index (§0.4); rename in one pass, not half the draft
21	P_4 **on both sides of the bridge**	Latent refactor hazard	TI_{m_1} must be the correlation conjunct only; comment it in the Alg. 5 implementation (§3.5b)
22	**Necessity overclaim at** m_2	**Retracted in §3.4**	Route A (definitional) or Route B (pointwise + measured margins δ); never "sound rejection by construction"
23	**GM described as hard AND**	Corrected in §3.3	GM = soft AND (zero-annihilating); hard gate = per-property UCB thresholds; always report the vector
24	**Underpowered null read as negative result**	Framing risk	Pre-register SESOI + equivalence test (§3.6); otherwise the result is inconclusive, not negative
25	**Screen and surrogate validated with one metric**	Conflation	Separate metric suites (§3.6); report decision utility (human ratings saved at fixed false-pass tolerance)
26	**Human margins** δ **on ladders never measured**	Route B currently assumption-only	Small human collection on perturbed ladder items; also settles whether ALPHA is human-invariant
27	**Criterion drift across human pools** ($\rho \approx 0.13$)	**Documented cause of the exp43 held-out null**	One rubric + one pool per claim; never pool the calibration panel with the 5-rater set; report the 0.13 as a construct-stability finding
28	**Single-rater criterion** ($\lvert H\rvert = 1$ on the panel)	Silent in the paper	State it; frame via $h_j = (\text{person}, \pi)$; inter-person ceilings only from the 343-label pool
29	**pass1/pass2 inverted vs. chronology**	Sha-verified trap	Rename to strict/lenient everywhere

#	Threat	Status	Mitigation
30	**MAE/MSE duplicate-handling convention mixed** in `main.tex`	Concrete instance of threat #10	Pick mean-of-duplicates; regenerate from one artifact run
31	**Audit folder uncommitted**	Pointers dangle for the team	Commit & push `experiments/26_human_label_dataset_audit/`

Part VII — Open questions

1. **Is TI_{m_1} aggregated over the persona-matched subpanel, or over the whole panel?** The definition demands the former. The experiments, as far as the notes show, may have used the latter. This is checkable and probably explains a sign flip.
2. **What is the right tolerance $\tau_{\mathbf{id}}$ for P_1?** The α-renaming variant behaves differently from `RENAME/REORDER/STYLE` (Fig. 13b isolates it for exactly this reason). Should α be its own property rather than a variant inside P_1?
3. **Does P_3 need the must-use-the-assumption guard?** In the VeriBench/BEq direction, a weaker spec can be trivially satisfiable; without the guard, P_3 has a precision problem.
4. **Should TI be reported per-agent as well as per-judge?** Algorithm 4 produces $\mathrm{TI}(aj \mid a')$. Its variation across agents is itself a certificate-robustness statistic and may be more interesting than the pooled scalar.
5. **What is the certificate's validity horizon?** A judge certified today on agent population A has a certificate that must expire as A drifts. Is there a re-certification trigger?
6. **Persona sample complexity.** How many scores n in D_{h_j} are needed before $\mathcal{M}[b, L_A, D_{h_j}]$ produces a judge that is TW_2 w.r.t. the *true* h_j and not merely w.r.t. the sample? The page assumes $n \approx 10$. This is the most promising place in the whole programme for an actual theorem (§1.6a).
7. **What does $\mathcal{M}$ actually do?** `38704` fixes $\mathcal{M}$'s signature but not its body. Candidates: rubric synthesis from D_{h_j}; few-shot conditioning on D_{h_j}; prompt search maximizing TI_{m_1} on D_{h_j}; fine-tuning. Each has a different overfitting profile w.r.t. n, and prompt search over D_{h_j} reintroduces exactly the selection circularity the property lock was meant to close.
8. **Is T_2 or T_3 the target?** (§1.6.2.) The existing human data is panel-with-shared-rubric, i.e. T_3. The definition as written is T_2. Decide, and if T_3, replace $\approx_\Pi$ in Definition 1 with an explicit aggregation operator.
9. **What is $\rho_{\max}$? — REVISED (§1.8.2):** not computable on the calibration panel ($|H| = 1$). The 0.249 inter-pass figure is single-rater *rubric* stability. Compute inter-person ceilings from the 343-label 5-rater pool, per rubric.
10. ~~How were the annotated $\hat{y}$'s selected?~~ **RESOLVED (§1.8.2):** stratified 5-per-split, hand-picked within split, frozen 2026-04-07 (`3a7b75c`), no script, no seed, not compile-filtered. Estimand: alignment on a fixed curated panel — say so in the paper; study-41's seeded template is the upgrade path if a reviewer demands a probability sample.
11. ~~Crossed or nested assignment?~~ **RESOLVED (§1.8.2):** fully crossed, trivially — one rater × 75 pairs × 2 rubric rounds. Inter-rater quantities do not exist on this panel; the 343-label 5-rater pool is where they live.
12. ~~Who is the third leaderboard agent?~~ **RESOLVED for the annotated slice (§1.8.2):** codex / claude / llama (`AGENT_MAP`). Leanstral and Qwen on `39218` were judge families, not annotated agents. (The *aspirational* ~10-agent leaderboard remains future work.)
13. **Does the bridge strengthen without P_4?** Report PTI_p and $\mathrm{PTI}_{p\setminus\{4\}}$ side by side (§3.5c). Costs nothing; data already collected.
14. **Is TI_{m_1} implemented as the correlation conjunct only?** Verify in code that no variance/stability term leaked into the human-side quantity.

15. **Subscript discipline in the draft.** One pass over the manuscript checking every TI/PTI subscript: m on the gold side, p on the proxy side, never swapped (the `TI_p` slip on the ratification page shows how easy the swap is).
16. **Route B constants.** Estimate human margins δ per rung/transform from a small ladder-rating collection; state ε; report which observed reversals exceed $2\varepsilon - \delta$ and therefore falsify pointwise alignment for that judge.
17. **SESOI.** Pick and pre-register the smallest deployable bridge correlation ρ_0 (candidate: the value at which the screen's human-savings at tolerance α drops below the cost of running it).
18. ~~Verify the claimed "Trust Index" title collision~~ **RESOLVED — collision is real.** arXiv:2507.16028, "From Logic to Language: A Trust Index for Problem Solving with LLMs" (Rug, Böhmer & Pfattheicher, Jul 2025). Consequence: TI_m remains internal notation; the paper's **title and public branding must not use "Trust Index."** Already consistent with the title shortlist.
19. **72-fix fork:** decide Option A (reword) vs Option B (rename 3 rows + regenerate) — recommendation B, see §1.8.3.
20. **State the $b' \to b$ extrapolation** (item-level generalization axis) explicitly in the paper (§1.8.1).
21. **Re-run the audit scripts before editing `main.tex`** — agent-reported numbers get the same verification discipline as any other numbers.

Part VIII — Paper packaging: title, protocol name, contribution language

Protocol name: amortized judge validation (screen–then–validate workflow). **System/repo brand (optional):** JudgeProbe is fine; zero urgency.

Title shortlist, merged and graded (constraint from #18: no "Trust Index" in any title):

#	Title	Source	Grade
T2	The Surrogate Gap: When Do Label-Free Behavioral Tests Track Human Agreement for LLM Judges?	mine + GPT's "surrogate" substitution	**9**
T7	Do Behavioral Stress Tests Predict Human Agreement? Scalable Validation of LLM Judges for Formal Code	GPT (question-form)	**9**
T8	JudgeProbe: Amortizing Human Validation of LLM Judges with Falsifiable Behavioral Tests	GPT	8.5 — needs the subtitle to carry the science
T9	Screen, Then Validate: Scalable Oversight of LLM Judges	GPT	8
T3	Cheap Screens for Expensive Judgments	mine	8 — better as talk title/subtitle
T10	Falsifiable Behavioral Properties for Amortized Validation of LLM Judges	GPT (conservative)	7.5 — safest, least memorable

#	Title	Source	Grade
T1	Necessary but Not Sufficient	mine	withdrawn until Route B is proven or Route A adopted

Candidate contribution block (GPT's four-contribution language, adapted to ratified notation — survives positive, negative, heterogeneous, and family-restricted bridge outcomes):

1. We distinguish scoped judge trustworthiness TW_m (construct), its human-grounded criterion TI_m (measured on a specified panel and item distribution), and the automated behavioral instrument $\mathbf{P}(aj)$ used to approximate it.
2. We introduce a reusable suite of falsifiable behavioral probes for LLM judges of formal-code artifacts, with pre-registered ladders, transforms, and estimators.
3. We evaluate whether $\mathrm{PTI}_{\{1,2,3\}}$ constitutes a valid surrogate for TI_{m_1} on held-out judge configurations, explicitly separating property violations (screen) from quantitative prediction (surrogate), with SESOI-anchored equivalence analysis where power is limited.
4. We measure the decision utility of the screen: human validations avoided at a fixed false-pass tolerance, with sensitivity to shift and a stated re-validation horizon.

One-sentence thesis: *We test whether automatically measured behavioral properties can predict a scoped human-alignment criterion for unseen LLM judges, thereby amortizing — not eliminating — the cost of human validation.*

Appendix A — Page-by-page transcription

A.0 `38700.jpg` — the definition page (written on a plane)

```
Definition (my proposed one) for Trustworthy AI Judge.
                                        → Solvers usually ai agents e.g. Codex.

For all Benchmarks (∀b in B), it should work for all agents or the subset of agents
            [above "ai judge" : ai judge → be "trustworthy"]
on the leaderboard (∀a in A or a in L_A, L_A ⊆ A), if the ai judge matches in
persona (or person) (e.g. strict judge, nice judge / non-strict etc) to the human
judge, then the (i.e. if aj = h_j / aj ≈ h_j) then the two judges should "match"
when they judge/score how well agent a did on solving tasks from benchmark b
(e.g. task could be: given question/data point x produce ŷ = a(x) s.t. ŷ ≈ y*
or just a good ŷ):

┌──────────────────────────────────────────────────────────────────────┐
│                          ∀x,y* in b        y* (optional)             │
│                              ↓      v                    ^ "match over" │
│ ∀b in B, ∀a in A or L_A, if aj ≈ h_j then                    i.d.x.    │
│     || P_θ(s | x, y*, ŷ = a(x), aj)  ≈_M  P_h(s | x, y*, ŷ = a(x), h_j) || │
└──────────────────────────────────────────────────────────────────────┘

 Def trustworthy ai judge            ↪ P_θ(s|…, aj) ≈_matches P_h(s|…, h_j)
                                       (1) m = matching criterion.

Options for "matching":  ┌ we want a method that produces such judges e.g.
                         | M_0(B, A, h_j) = aj  or  M_1^s(b, a, h_j) = aj
                         | or M_2^s(b, A, h_j) = aj  or  M_2^s(b/A, h_j) = aj
                         └                              ↑ want to condition on less

0) Strongest/hardest: ai judge produces the exact distribution as the human
   person in question.
     ↪  P_θ(s | x, y*, ŷ = a(x), aj)  =  P_h(s | x, y*, ŷ = a(x), h_j)
                                      or ≈

1) Most common: ai judge correlates with human judge: e.g.
```

```
   Corr( D^s_aj = {s ~ P_θ(s | x, y*, a(x), aj)}^{N^s_aj} ,
         D^s_hj = {s ~ P_h(s | … h_j)}^{N^s_hj} )

   Corr( aj scores correlate to hj scores ) = Corr( S_aj , S_hj )
                                                  └──────┬──────┘
   where corr can be Spearman, R², Kendall,          || S_aj ≈_corr S_hj ||
   Spearman, ICC, etc.

2) My favorite modification to 1: add stability, e.g. aj is low variance/std
   So  Corr( S_aj ≈ S_hj )  &  Var_{P_θ(s | x, y*, a(x), aj)}[ S_aj ] ≈ low variance
```

Partially visible on the sheet beneath: ∀b ∈ B, ∀… / "This exam[ple] … that hu[man] judge … scores …" — appears to be an earlier draft of the same statement.

A.0b `38704.jpg` — method disambiguation and the h_j pointer problem

```
TODO: Next task: Clarify what M_i's mean especially
M_2, M_2', M_1  +  What/Conditioning on hj means (e.g. we usually have a
human we can ask labels for).
                        → P_θ(s | …, aj) ← e.g. [prompt, etc. that]
                                           leads [to] def TW.

M_i is a method that outputs a judge with the Def of trustworthiness that
we want in (2)  [scribbled] / Def TW_i (Trustworthy).

assume we have hj e.g. a strict judge (e.g. Rylan) / persona.
So the method could take this person and [scribbled out] output
a judge aj st it has property/cnz Def TW_i  e.g. [scribbled] TW_2
(since it's my favorite one).  (actually doable in practice, I think)
+ has nice conditions → matches/corr human in a stable way).

=> Let's start with method 2'  M_2' .  So for method 2 an ai
judge works/TW  (works ≈ trustworth[y] a la def TW_2 [scribbled] for example)

┌──────────────────────────────────┐   So fix a benchmark e.g. VB, fix a
│ Method_2' = M(b, L_A, hj) = aj │   leader board  L_A ⊆ A  and
└──────────────────────────────────┘

a human hj (strict person or person Rylan).  But in practice we don't
have a pointer like this one to the person/persona.  Instead we
usually have scores/labels/judgments from that human
judges/target judge(s).  e.g. 10 scores according to the persona/perso[n]
or person + persona ≈ persona (overloading terminology).

    model/agent }   prompt / θ /
    identity    }   person

[bottom, partially cut off at the page edge:]
or even a collection of scores from different humans.  In the
past all the humans would get the same prompt/rubric
[for] judg[ing] and produces judgments & we want an aj / Method that
outputs a good TW_2 judge
```

Reading notes. The circled (2) refers to option 2 on `38700.jpg` (correlational + stability), confirmed by *"since it's my favorite one"*, which echoes *"My favorite modification to 1"* on that page. The top-right marginal annotation is partially illegible; it appears to gloss $P_\theta(s \mid \cdots, aj)$ as being determined by the prompt and whatever else the definition of TW fixes `[?]`.

A.0c `38707.jpg` — concrete instantiation of Method 2' (work in progress)

```
Example: Method_2'      b = VBv1 = VeriBench (v1)
                        ──────────────────────

L_A = some leader board for this method   L_A = { codex   ,  CC     , lenshal[?] , Qwen[?]
                                                 GPT5.6     Fable5                /gemini }
```

```
                                        ultra      max
                                        ___________________________________________

instead of hj we have    D^S_hj = { s ~ P_h(s | x, y*, ŷ = a(x)) }^{N^S_hj}

or if we have a set of humans H:   D^S_H = { ∪_{hj in H} { s ~ P_h(s | …) } }   [scribbled]
                                     ↪ H = { Ethan, Yegor, …, etc. }

      5.10    5.5  = 5(10+5)
      50+25          5(15) maybe?   in each judge
                                    scores

D^S_H = e.g. 75 = 5 people x 15 judgments  across  random ŷ = a(x)  &
                                                    ^__________┘
selected random ŷ's.

Q: Ok I don't remember how [→ arrow to "ŷ = a(x)"] ŷ's were selected. But had
issues with leaderboard[?] or something. So let's figure this out so I can do this
clearly.
```

Reading notes. `CC / Fable5 max` reads as Claude Code running Claude Fable 5; `codex / GPT5.6 ultra` as Codex running GPT-5.6 ultra. The third leaderboard entry is illegible `[?]`; the fourth reads as Qwen and/or Gemini. The word after *"had issues with"* is illegible — best candidate is "leaderboard", but it is not certain `[?]`. The blue arrow connects *"selected random ŷ's"* back to $\hat{y} = a(x)$ in the line above, i.e. the concern is about which agent outputs got sampled for annotation.

A.0d `20260812_1324.jpg` — the rename

```
What we really want is a Trust Index that predicts the trustworthiness
                        ___________ [circled / scribbled over]
score.
          ↪ the term for our certification.

→ I think it should be renamed cuz its confusing re
  we have TW_i, TW_2, TW_2' etc. Those [scribbled] are
  better named as the trustworthy indices info.[?]

    ↪ and our index as the Certification Index.
                          ____________________
```

Reading notes. The word after "trustworthy" reads "indices"; the trailing token is either "info" or "inho" `[?]`. "Trust Index" in the first line is underlined and then scribbled over, which is what the rename proposal is reacting to. Acted on in §0.4.

A.0e `20260812_1348.jpg` — the ratification page (final notation)

```
Def Trust Worthiness_m  (TW_m)   (circled mark)
_____________________________

∀b in B, ∀a in A or L_A,  if aj ≈ h_j then
       ^ ∀x,y*
   || P(s | x, y*, ŷ = a(x), aj)  ≈_m  P(s | x, y*, ŷ = a(x), h_j) ||_m
                                                        "in 'm'"  e.g.

↪ trustworthiness in the sense matching ? in "m" e.g.
   m can be * m_0 = distributional;  P(s|…,aj) ≈ P_h(s|…,h_j),
   m_1 = most common / correlate with human judgments;  rho := Corr(ŝ_aj, ŝ_hj),
   m_2 = m_1 fav: correlates with humans & stable:
                 ≈ Corr(ŝ_aj, ŝ_hj)  and  std(ŝ_aj) ≈ low
                                        (Var)
↪ TW_2  or  TW_m2   (notation).

So an AI Judge is trustworthy in the sense of TW_m if [?]
its Trustworthy score is high (the proxy … we really want [?]).

I ↪ [?]  → PTI_p
```

```
        → TI_p.

→ The goal of PTI_p => if it's low  => falsifies/rejects ai Judge.
                                      approximately [?] / approx
                  => if it's high => "certifies" the ai judge – cuz
                     we demonstrated that PTI
       ———→  does Correlate with TI [check]!
             (underlined: does)
             key!
```

Reading notes. `TI_p` in the middle is read as a slip for TI_m — p indexes properties, which exist only on the proxy side (logged, Appendix C). The marginal "approximately" over the rejection arrow concedes that rejection is statistically approximate; the scare quotes on "certifies" concede that acceptance is bridge-conditional. Both are the correct semantics and both were adopted (§0.4, §3.4).

A.0f `39217.jpg` — TI_2' tensor form and the boxed formula

```
[scatter sketch] ŝ_aj scores (0.0→1.0, vertical) vs ŝ*_hj (horizontal); one x point.

[red] Q: ok I think I clarified some parts. What is still not fully clear is how
VB's & cert Judge's current paper/data map to all these params/sizes. e.g. WHAT IS K

i_b = 1 like I think it has to be on k same examples:
 → S = [ [Ŝ_aj(i_b, i_a, i_k)]_{i_k=1}^{K} , Ŝ*_hj(i_b, i_a, i_hj) ]

for simplicity say k = 1, but we could try correlating all the samples of the
ai judge (aij) score with that "single" human attempt.
 ↪ note we can also produce lots of ai judge scores lol. (same for human).

S = [ [Ŝ_aj(i_b,i_a,i_ka,i_kaj)]_{i_ka,i_kaj} , [Ŝ*_hj(i_b,i_a,i_hj,i_khj)] ]
D_{Ŝaj,Ŝ*H} = D_Ŝaj [|b|x|L_A|xK_axK_aj]  x  D_Ŝ*H [|b|x|L_A|x|H|xK_hj]

Cool so we correlate  Corr(D_Ŝaj, D_Ŝ*H)                    }
& maybe sure  STD_{P_θ}[D_Ŝ*H [?aj]] ≈ low.                  } TI_2'
  ┌──────────────────────────────────────────────┐           }
  │ TI_2' = (Corr(D_Ŝaj, D_Ŝ*H) * STD_{P_θ}[D_Ŝ*H])^{1/2} │
  └──────────────────────────────────────────────┘
 ↪ this is what we want to approximate with PTI
```

Reading notes. The STD subscript glyph is ambiguous between H and aj; P_θ types it as the judge's own scores (see §1.7 corrections: orientation flip, subscript, clipping).

A.0g `39218.jpg` — TW_2' + the data-size question

```
Goal: PTI be "equivalent" [to/when using] the "gold/true" TI.
TI_1 ≈ usually human correlations. But I have a better definition imho:
TI_2 or (TI_2') → TI_2' := [ai judge] correlates to human judgens & is stable –
  [* low std] – for all agents/solvers in the Benchmark in question.

TW_2' := b, & ∀a in L_A, if aj = hj then ∀x,y*, Corr(Ŝ_aj ≈ Ŝ*_H)
                          [orange: Q TODO: clarify what this means]

Ŷ = [|b| x |k| x |L_A|] samples
Ŷ = [|b| x |L_A|] ≈ 800ish x 10ish   (# problems in benchmark x # agents/solvers
b = VBv1                              in leaderboard) ↪ could have multiple
                                      samples… but for now 1.
      | CC | Codex | Leanstral | Qwen |        [margin:] Ok let's think. What
  CS  | ŷ  |  ŷ    |    ŷ     |      |        that means. What data do we have
  hE / ES / SS / RCS / IS rows          for the set of humans H = {h₁,h₂…}
  [diagonal line across table]                = {Yegor, dan…[?]}

→ each of those |b| has a single gold ref ŷ* answer
 ↪ Ŝ*_H = [|b| x |L_A| x |H|] = Ŝ*(i_b, i_a, i_hj)   [margin: index, agent/s idx]
 ↪ this IS what [it] should mean! Have Human judgments tensor/data Correlated
   with the aj (ai judge etc)   Ŝ_aj = [ …   [cut off]
```

A.0h **39296.jpg** — effective parameters and b' ⊆ b

```
Ok what did I tell Rylan in the car?     We want to approximate TI(b, L_A, aj, k_aj, H, k_H)
  b' ⊆ b           * effective params: (b, L_A, aj)              (b, L_A, H)

TI(b, L_A, aj, k_aj, H, k_H) <-> TI( D_Ŝaj [|b|x|L_A|xk_axk_aj] , D_Ŝ*H [|b|x|L_A|x|H|xk_H] )
┌──────────────────────┐             |b'|                          |b'|
│ TI( b', L_A, aj, H ) │ [care!]
└──────────────────────┘
```

note: => my notation with |b| assumes we got agent/solver generations & their ai judge scores for all b (e.g. VB) – but really I/we only did b' ⊆ b a subset of VB/b of course; duh cuz human evaluations are expensive. That's the whole point of PTI; to replace the human judgment necessity as much as possible.

↪ we just want basically to establish ||aj ≈_M H||, ai judge matches scores of human pool H = {yegor, dan[?], ethan, ... etc.}

```
        D_Ŝaj          v cuz we have "one entity/judge" we want to establish
          1_x            is trustworthy (or not) cert H          D_Ŝ*H
[|b'| x |L_A| x k_a x k_aj] , [|b'| x |L_A| x k_a x |H| x k_H]
[|b'| x |L_A| x k_a x 1 x k_aj] , [|b'| x |L_A| x k_a x |H| x k_H]
```

A.0i **39297.jpg** — effective params of PTI

```
Q: What are the effective params of PTI (Proxy Trust Index)?
→ def the benchmark b (e.g. VB)          "proxily"/approximately.
→ def the ai judge (aj) we are "certifying".

→ PS: I suppose in practice we can only run a subset of solvers/agents for the
benchmark b e.g. L_A but since PTI is "scalable" (somehow; to be specified) then
L_A can actually be large (?) [I think since the properties P alg is do-able
scalable/automatic thing; once P has been specified].

→ And def the properties P themselves         ^ I feel reflexivity is too
  (reflexivity → equality? (to be renamed))     mathematical, hard to explain
  (reflexive, monotonicity-bugs,               simply to a lay person, e.g. a VC
   mono-specs-missing, ...)                    or even Rylan in the car; he would
(are arguments) (k_x is [might be prompt,      have asked me what reflexivity
 but] it's distracting me right now).          means.

PTI( b , L_A, aj, P )  ->  s.t. ≈  TI_2' ( b', L_A, aj, H )
     ↑ full VB possible!           e.g. correlates
(TODO: don't rly know yet if that's a good idea).
TODO: after break; how do you get D_S's for PTI [written "PIT"]?
```

A.0j **39399.jpg** — Def Trust Worthiness 2' (arguments form)

```
Def Trust Worthiness 2')
                                        Corr( Ŝ , Ŝ* )
TW_2'(b, aj, L_A, H) = Corr( aj(b, L_A(b)) , H(b, L_A(b)) )   *
                         └──── Ŝ ────┘   └──── Ŝ* ────┘
                     STD( aj(b, L_A(b)) ) [check]
                     └──── STD_aj[ Ŝ ] ────┘

TW_2'(b, aj, L_A, H) = ( Corr( Ŝ_aj , Ŝ*_H | b, L_A )  *  STD[ Ŝ_aj | b, L_A ] )^{1/2}

┌─ Def TW 2) ─┐
TW_2'(b, aj, L_A, H) = ( Corr( Ŝ_aj , Ŝ*_H | b, L_A ) * STD( Ŝ_aj | b, L_A ) )^{1/2}
```

A.0k **39398.jpg** — dataset form, candidate pool L__A(b')

```
TW_2'(b, aj, H, L_A) = [scribbled out]
   = Corr( aj(b, L_A(b)) , H(b, L_A(b)) )      and low variance
           └──── Ŝ ────┘   └──── Ŝ* ────┘
   = ( Corr( aj(b,L_A(b)), H(b,L_A(b)) ) * STD( aj(b, L_A(b)) ) )^{1/2}
```

```
Ok now in terms of D_S etc.      → L_A leaderboard.. we want the judge to work on.
Think first before checking the agreed notation from yesterday.

  x,y* in b      a = ŷ(.) in L_A
→ aj( x, y*, ŷ=a(x) ) = Ŝ( x, y*, ŷ_a(x) | aj ) = Ŝ_aj( x, y*, ŷ )

┌──────────────────────────────────────────────────────────┐
│ Ŝ_{x,y*,ŷ_a|aj} = Ŝ_aj( x, y*, ŷ_a(x) ) = aj( x, y*, ŷ_a(x) ) │
└──────────────────────────────────────────────────────────┘

• Consider x,y* in b then [scribbled] we want (WW)[?]     aj ≈_{TW_2} H more
  Precisely   ┌──────────────────────────────┐   Goal of when one wants to
              │ Ŝ_{aj|b,L_A} ≈_{TW_2'} Ŝ*_{H|b,L_A} │   generate trustworthy judges.
              └──────────────────────────────┘

       D_{Ŝ_aj|L_A(b)} = { Ŝ_aj(x, y*, ŷ_a(x)) }_{x,y*,ŷ_a in L_A(b')}
Corr( D_{Ŝ_aj|b',L_A} , D_{Ŝ*_H|b',L_A} )
       → D_{Ŝ_H|L_A(b')} = { Ŝ*_H(x,y*,ŷ_a(x)) }_{x,y*,ŷ_a in L_A(b')}

L_A(b') = { (x, y*, ŷ_a(x)) | a in L_A },   L_A' ⊆ L_A [?],   b' ⊆ b

– Select some subset of data points from b, b' ⊆ b to give to your agents/solvers
  & try to have aj's scores to match TW_2' to H's.

Corr( D_{Ŝ_aj|L_A(b')} , D_{Ŝ*_H|L_A(b')} )
```

Reading notes. The boxed goal's `Ŝ*_{A|b,L_A}` is read as `H` (A is the agent set; it would be a type error there). Two adoptions from these pages: the **candidate-pool operator** $L_A(b')$, which makes pairing a typing fact rather than a side condition, and the **conditioning-bar dataset notation** $D_{\hat{S}_{aj}|b',L_A}$. Two corrections: the orientation flip is still missing $(1-\mathrm{STD})$, and $\mathrm{TW}_{2'}(\cdot) = (\mathrm{Corr}\cdot\mathrm{STD})^{1/2}$ defines the *scalar*, i.e. $\mathrm{TI}_{2'}$ — TW is the predicate $[\mathrm{TI}_m \geq \tau]$.

A.0l `39404.jpg` — PTI as GM over properties; the boxed bridge target

```
P_{rfl, agg_rfl} = Agg_rfl( { Ŝ_{aj,x,ŷ,y*,perm}(alpha, alpha) }_{alpha = all x, y*, ŷ +
                                                    any perturbations perm} )

→ then  PTI(b, L_A, aj, P) = "smooth" conjunction of all properties. So
                             e.g. GM of all agg's scores.

PTI = [ P_{rfl,aggrf}( D_{Ŝ_{aj,x,ŷ,y*,perm}} ) * P_mb(..) * P_mms(...)
                                          * P_jsar[?](...) ]^{1/# properties}
                                                          ↓  |P|

P_mb( Ŝ ) = insert bugs monotonically.  Corr should be negative.
                                       ↪ if negative the |mb(.)| => high, so PTI is high.

=> then given PTI we want to make sure etc      } for any properties.
   we can achieve PTI s.t.

┌──────────────────────────────────────────────┐
│  PTI( b, L_A, aj, P )   ≈_m   TI_2'( b, L_A, aj, H ) │
│                          ↓  Corr,                    │
│              TW_2' = Corr & stable/low std           │
└──────────────────────────────────────────────┘
```

A.0m `39403.jpg` — PTI motivation, property names, P_rfl over three slots

```
Ok now that the "true" trustworthiness score is clear let's figure out the PTI
motivation too.

[TW_2' crossed out](b, aj, L_A, H) = ( Corr( D_{Ŝ_{x,y*,ŷ}|aj, L_A(b')} ,
TI_2'                                      D_{Ŝ_{x,y*,ŷ}|H, L_A(b')} )
                                    * STD( D_{Ŝ_{x,y*,ŷ_a}|aj, L_A(b')} ) )^{1/2}
```

```
Ok now  PTI = PIT(b, L_A, aj, P)   note: b is whole b not b'!!
                                ↪ instead of humans. the hypothesis is that
The judge should have properties. If it has these properties then it (PTI) this
proxy metric correlates/matches (e.g. using) TI_2' just cuz it just means
"correlates" + low std/variance).
               bugs   missing specs     prompt stability & semantic invariance
P = { P_rfl, P_monob, P_mono-ms, P_ps&SI } = { P_r, P_mb, P_mms, P_ps&SI }
                                                        ↪ split it into two.


P_psSI = [crossed out]
It's more  Judge stability to prompt perturbations via Semantic Invariances
(a little too broad wording, relative to what we really did.)

P_rfl = judges scores don't change when the files are the same
 ↪ ≈ Ŝ_{aj,x,ŷ,y*|Prf}( (x,x), (ŷ_a,ŷ_a), (y*,y*) ) ≈ 1.0  ∀ all 3.
```

Reading notes. Three corrections carried into the memo: $|\rho|$ is the wrong orientation map for P_{mb} (it also rewards $\rho \to +1$, a maximally anti-monotone judge) — use $\frac{1-\rho}{2}$; the boxed target should read $\text{TI}_{2'}(b', ...)$, since the gold is only measurable on the annotated subset (as the second page itself notes); and the $\approx$ between PTI and $\text{TI}_{2'}$ ranges over **judge configurations**, one level above the item-indexed $\approx_m$ inside TW_m. The page's own annotation "instead of humans" is the thesis: **P** occupies the argument slot of **H**.

A.0n `39659.jpg` — TW_2' spelled out in words; the cost model

```
Goal: have a way to replace the process of using human evals to establish the
trustworthiness of of an ai judge.  TODO: need to make clear the motiv/cost analysis.
This: TW_2' in particular ( for now

TW_2'(b',L_A,aj,H) = [ Corr( Ŝ_{aj,L_A(b')}, Ŝ*_{H,L_A(b')} ) * (1 - std( Ŝ_{aj,L_A(b')} )) ]^{1/2}
                                                                    └ leaderboard
Ŝ_{aj,L_A(b')} = "scores of aj (ai judge) on (gen) data produced by agents in L_A
                  with benchmark subset b'!"
Ŝ*_{H,L_A(b')} = "scores of collection of humans H of ans/data produced by
                  agents/entities/ai's in leaderboard L_A."
              = { Ŝ(x, y*, ŷ_a(x)) }_{x,y* in b', ŷ_a(.) in L_A}

Ŝ_{aj,L_A(b')}  = D_{Ŝ_aj,L_A(b')}  = { Ŝ_aj(x,y*,ŷ_a(x)) }_{x,y* in b', ŷ_a(.) in L_A}
Ŝ*_{H,L_A(b')} = D_{Ŝ*_H,L_A(b')} = { Ŝ*_H(x,y*,ŷ_a(x)) }_{x,y* in b', ŷ_a(.) in L_A, h in H}
     ||  { Ŝ_h(x,y*,ŷ_a(x)) }_{x,y* in b', ŷ_a(.) in L_A, h in H}


┌──────────────────────────────────────────────────────────────────────────┐
| TW_2'(b',L_A,aj,H) = [ Corr(Ŝ_{aj,L_A(b')}, Ŝ*_{H,L_A(b')}) (1 - std(Ŝ_{aj,L_A(b')})) ]^{1/2} |
└──────────────────────────────────────────────────────────────────────────┘


the expensive thing usually is the "production" of hum labels, so Ŝ*_{H,L_A(b')}
(oh oh assuming the benchmark already exists...)

O( |H| . |b'| ) = per human we pay 1 human judgment "cost" per triplet but we score.

note: |b'| < |b|   (<<)
```

Reading note. First page in the stack with the orientation flip written correctly $(1 - \text{std})$. The cost model needs one more factor: $O(|H| \cdot |b'| \cdot |L_A|)$, since each problem contributes $|L_A|$ candidate triplets — which reproduces the audited 144 labels exactly ($1 \times 24 \times 3 \times 2$ rounds).

A.0o `39658.jpg` — PTI goal, property names, P_rfl aggregation

```
Goal: the goal of PTI (Proxy trust index) is to approximate TW_2' ( in this case
PTI_2' ≈ TI_2' ).

TW_2'(aj,L_A,b',H) = ( Corr( Ŝ_{aj,L_A(b')}, Ŝ*_{H,L_A(b')} ) * std( Ŝ_{aj,L_A(b')} ) )^{1/2}

PTI_2'( aj, L_A, b , P ) = TODO write what this equation really means.
                ↑ [b, not b']    ↪ Q: product of property metrics?
```

```
  but we [can] use full benchmark cuz there is no human judgments; or the cost is
  to produce the properties once & verify why do Correlate with TW_2' ie by lead
  to TW_2' ≈ PTI_2'.

P = { P_rfl, P_mb, P_mms, P_jsI }  → P_jsI = Judge Stability to alpha-Semantic Invariance
  { "     "     "    ,  P_alphaSI }      (ie prgms, renaming bound variables).

. P_rfl  we can set all of  x, y*, ŷ_a(x)      [diagrams: x,y* in b column;
                            inb   in bxL_A        ŷ_a(x) in b x L_A grid]
 ↪ conceptually, for any two files that are the same, the judge says they are the same

so ┌──────────────────┐ goal.   D = some data = { (y, ŷ) }_{y in D}
   │ P_rf( D, aj ) ≈ 1.0 │        ↪ but this could be anything eg we can introduce
   └──────────────────┘             bugs in y* or ŷ(x) and see if judge can still
   or x given too if this           say the two files are the same.
   judge takes x for same reason.

 ↪ ┌──────────────────────────────────┐ → ideal ≈ 1.0
   │ prod_{y in D} Ŝ_aj( y, y ) = P_rf( D, aj ) │
   └──────────────────────────────────┘

      ↪ oh this is the aggregation metric [that] I were talking about in the paper.
   PS: exponential penalty if lots of these aren't 1.0. I guess for such a single
   property this is fine.
```

A.0p `39657.jpg` — P_mb estimator and the bug-ladder picture

```
P_mb( aj, ... ) = monotonicity to bug insertion. ie. as a bug ladder is introduced
       the scores of the judge goes down monotonically & we can aggregate the
       correlations of these correlation/bug insertions. e.g.

P_mb( aj, D, bugs ) = prod_{y in D} ( 1 - Corr( Ŝ_{aj,bugs}(y) , bugs ) )
          ↑              || {b_0, b_1, ..., b_{#bugs}}
  - say gold ref            ↪ D_{L_A(b)} or D_b        ↪ for each file we have a
  - or outputs of some set of agents L_A(b)              correlation of the bug
                                                         ladder to their judges aj
Ŝ_{aj,bugs} = { ( b_i , S( b_i(y), y ) ) }               score.

[graph: judge score decreasing across bug ladder b_0, b_1, b_2, ..., b_{#bugs}]
                        → for 1 file so for just one  ŷ, y ; S( ŷ, y )
Corr( Ŝ_{aj,bugs} , bugs ) ≈ negative            e.g. S( ŷ, y* ) or S( b_i(y*), y* )
So   1 - Corr( Ŝ_{aj,bugs} , bugs )                  or S( b_i(ŷ), ŷ )
```

Reading note. Two corrections carried into the memo: raw products are scale-dependent (72 items at 0.99 give 0.48; the same judge on 471 candidates gives 0.009), which would make PTI on b and TI on b' incomparable by construction — use the geometric mean $(\prod)^{1/|D|}$ at both aggregation levels; and $1-\rho$ spans $[0,2]$, so the map must be $\frac{1-\rho}{2}$.

A.0q `39969.jpg` — the bridge plot sketch and the two anchor judges

```
TW_2'(aj, H; b, L_A) = [ Corr( Ŝ_{aj,L_A(b')}, Ŝ*_{H,L_A(b')} ) * (1 - std( Ŝ_{aj,L_A(b')} )) ]^{1/2}
                            ||                          ||
                  { (ŝ_aj(x,y*,ŷ(x))) }   { (ŝ_hj(x,y*,ŷ(x))) }

PTI_2'(aj, P; b, L_A) = agg P's.

TW_2' will tell us how trustworthy aj is wrt TW_2' metric.
  for b = VB, L_A = some leaderboard eg = {CC, Codex, Leanstral, Qwen, Gemini}
       → any judge, including random guessing/random snoozing[?] should score
         low hopefully. (sanity check)

← let's first see if PTI_2' correlates with TW_2'
 ↪ TODO: would be nice to figure out a way to visualize this just like we
   visualize two things correlating.

[Plot 1] y: TW_2' (target) 1.0 ; x: PTI_2' (proxy) 1.0 ; x marks on a rising line
```

```
→ X_UR = ( TW_2'(aj = random uniform, H) , PTI_2'(aj = random uniform, P) )
           └ 0.1 * (1-0.9) = 0.1.0.1 = 0.01   └ say it says, 0.013, ie it's not
                    ~higher                     really trustworthy wrt properties P.
   ( this will omit to keep notation simple )

→ X_GAJ = ( TW_2'(aj = Good AI Judge, H) , PTI_2'(aj = GAJ, P) )
           └ 0.89                          └ 0.874

[Plot 2] same axes, x marks on rising line, plus a bell curve along the bottom
         Corr / STD      TI = 0.89

→ This is the right plot!
→ but maybe we can start with 2 figs eg in py matplotlib, & try to merge them
  somehow into this really cool plot.
→ Maybe even implement it in utils
```

A.0r `39970.jpg` — score data written out; GM superscript adopted

```
Goal: use properties to replace using human judgements.

TODO/Q: we use Human judgments to validate this method – so how do we make sure we
don't need to collect H's every time we have a new benchmark? later. But mostly
basically generalization. Otherwise it feels it's the same process except now we
have an extra step.

TODO: I think writing down the score data will really help understand my method.

For TW_2'(b',L_A,aj,H) = [ Corr( Ŝ_{aj,L_A(b')}, Ŝ*_{H,L_A(b')} ) * STD( Ŝ_{aj,L_A(b')} ) ]^{1/2}
                                          [blue correction:] ( 1 - STD( Ŝ_{aj,L_A(b')} ) )

Ŝ_{aj,L_A(b')} = { ( ŝ_aj(x, y*, ŷ(x)) ) }_{x,y*,ŷ(x) in L_A(b')}
Ŝ_{H,L_A(b')}  = { ( ŝ_hj(x, y*, ŷ(x)) ) }_{x,y*,ŷ(x) in L_A(b'), hj in H}

Want to write PTI_2'( aj, L_A, b , P ) = Agg property scores.
                                  ↑ full
P^GM_rfl( D, aj ) = ( prod_{y in D} Ŝ ...
```

Reading notes. The blue in-line correction confirms $1 - \mathrm{STD}$, and the `GM` superscript on P_{rfl} adopts the scale-free geometric aggregation. The open TODO/Q — whether a new benchmark forces a fresh human collection — is the $b' \to b$ and cross-benchmark generalization axis already registered in the memo; the honest current answer is that the bridge is validated within one benchmark and transfer across benchmarks is untested.

A.0s `PXL_20260817_190042317.jpg` — the M^VS operator; aggregator as argument

```
Really; all that's missing is the actual equation for PTI_2' (proxy trust index).

PTI_2'( aj, P ; L_A, b/D ) = some agg function of properties
                    or something like that... depending on the ai judge
                    e.g. if aj doesn't take x its fine =.

P^GM_rfl( aj, P_rfl ; D ) = ( prod_{y in D} Ŝ_aj(x, y, y) )^{1/|D|}   (GM).

 ↪ P^GM_rfl( aj, P_rf ; D ; GM )
            [core args]  [given "hyperparams"/meta/explan.]
   To be decided.

P^Agg_mb( aj, P_mb ; D, Agg )

 ↪ per y in D we get 1 scatter plot
           ↓
   Validity-Stability scatter plot / VS - scatter plot.
           or
   just traditional (corr) - scatter plot.   C - scatter plot

[diagram] Ŝ ↑ x                 VS( sigma, rho_c ) = M^VS_{rho_c, sigma}( aj ; D )
```

```
|   x x                      ↑ Ŝ_aj(x, y, y_b)
+--+--+--> bug ladder  [ sigma = std( Ŝ_aj ; D )
  b_1 ... b_5            rho_c = Corr( Ŝ_aj , D ) ]
                       ↪ in this context it's VS more than TI_2'.

VS( aj ; D ) = M^VS_{rho_c,sigma}( aj ; D ) = M^VS( D_scatter-plot-data )
             = M^VS( D_1 , D_2 )
                 v        ↘
                D_1        D_2
                 ↓          ↓
                S_1        S_2
             eg Ŝ_aj     bugs in this case.
```

Reading notes. The key move: $\mathcal{M}^{VS}$ is a two-argument functional, so $\mathrm{TI}_{2'} = \mathcal{M}^{VS}(\hat{S}_{aj}, \hat{S}^{*}_{H})$ and $P_{\mathrm{mb}} = \mathcal{M}(\hat{S}_{aj}\restriction_{\mathrm{ladder}}, \text{bug level})$ are the *same measurement against different ground truth* — the $\mathbf{H} \to \mathbf{P}$ swap at the operator level. Caveat carried into the memo: instantiating the full $\mathcal{M}^{VS}$ on both sides puts $(1-\mathrm{std}(\hat{S}_{aj}))^{1/2}$ into both scalars and makes part of the bridge correlation mechanical; properties therefore use the correlation-only component $\mathcal{M}^{C}$ and the target keeps the variance factor.

A.0t `PXL_20260817_193801298.jpg` — per-item VS aggregation; notation decision

```
Trying to figure out what  P^agg_mb( aj, P_mb | D | agg )  (o)

PS: I think I'm going to use Bayesian "given"/(condition) vs frequentist in my
handwritten notes... but in the paper we should use ; notation. It's just | is
easier to visually see.

→ for each data point we have  [ VS( Ŝ_aj,y,bugs , bugs ) ]_{y in D} = M^VS
   Ŝ_aj(y|b|y)                          D_1        D_2
   vs = corr & std
   [small scatter sketch] → bugs
                        ~ Corr( Ŝ_aj,y,b,bugs , bugs )  &  std( Ŝ_aj,- )

Q: how would we (aggregate)/combine these in P^agg_mb(.) (o) to get the "actual"
   property score.

→ Intuitively/first principles; What we want is; for each y in D ie for each
  VS-scatter plot of ( Scores vs bugs ) we want them to (negatively) correlate
  & scores be stable. This again sounds to me like a GM.
                                      [margin: y in D ~ perturbation wrt graph]

┌──────────────────────────────────────────────────────────────────┐
| P^GM_mb( aj, P_mb | D | GM ) = ( prod_{y in D} ( 1 +/-[?] Corr( Ŝ_y, bugs ) ) )^{1/|D|}
|                                                  { VS
|  M^VS = agg_mean[?]         = ( prod_{y in D} M^VS( Ŝ_aj,y,bugs , bugs ) )^{1/|D|}
└──────────────────────────────────────────────────────────────────┘

So I suspect this applies to all remaining properties, cuz for MS (missing specs)
it's the same. As more specs missing => judge should be "angrier"/lower score.
For Semantic inv. alpha-renaming it should be robust to all changes. So we want high
Corr/VS for all y in D
```

Reading notes. Four items carried into the memo. (i) The boxed sign reads as $1 + \rho$, which is inverted — at the ideal $\rho = -1$ it returns 0; the normalized form is $\frac{1-\rho}{2}$. (ii) The stability component inside a ladder property must be repeat variance at a fixed rung, never spread along the ladder, which should be *large* for a good judge. (iii) The page argues for $\mathcal{M}^{VS}$ on the property side, which flips the earlier recommendation: adopt **Config B** — gold is plain human correlation, properties carry the variance — matching both this formulation and the existing implementation. (iv) The unification covers the two ladder properties only; P_{rfl} and $P_{\alpha SI}$ have a constant second argument, so they take a deviation form $1-\mathbb{E}|\Delta s|$ rather than a correlation.

A.1 `37253.jpg` / `37251.jpg` — Trust Index definition sheet

```
let/fix, b = vB
∀a in A ~ L_A                    Find a j (ai judge with properties bellow:)
```

∀h_j in H(b), ∀x, y* in b

|| ŝ(x, ŷ, y*, r+aj) ≈_{Pc} s*(x, ŷ, y*, r+h_j) ||_{Pc} = Corr(ŝ, s*) = high
&
Geometric mean ^ low worse:
Var(ŝ, s*) = low.

b = vB, ∀a in A, ∀h_j in H(b), ∀x,y* in b, GM(Corr(ŝ,s*), Var(ŝ,s*)) = good
↓ Δ4
1 - std(ŝ,s*)

GM(Corr(ŝ,s*), 1 - std(ŝ,s*)) = high

high low → high
(higher is better).

Q | TODO: this seems an optimization over the judges. That's fine with me.
But, what we need is – at least per-cert-judge – a Trust Index that "works".
So human evals we done once. [margin: scalable
Generalization across a in A or that it (Scalable certification of llm/ai judges)
works in the given A, is really important. → at the very least for the leaderboard.]

→ So I think there are two papers
1) finding the "good judge" / certifiable judge
2) find j for (hopefully 1st's?) judge – Trust Index that removes from us
the need of human labels.

A.2 `37252.jpg` — rotated loose sheet (partial; low legibility)

∀a in A ~ L_A
find aj such that
|| ŝ(x, ŷ, y*, r+aj) ≈_{Pc} s*(x, ŷ, y*, r+h_j) ||_{Pc}
for any relevant human judge, i.e. ∀h_j in H(b). (note x, y* in b of course)

Said more cleanly:
for b = vB (let b = vB)
∀a in A ~ L_A (all agents in consideration)
∀h_j in H(b) – ai judge score should match relevant human judges' scores, i.e.
|| ŝ(x, ŷ, y*, r+aj) ≈ s*(x, ŷ, y*, r+h_j) ||_{Pc}
∀a in A's benchmark → "multiple plots" [→ a1, a2, a3 …]
e.g. Corr(ŝ, s*) is (unnormalized) [?]
→ low variance … styles etc.
for VB can't we [judge] style etc. for benchmark VB.

[right margin, partly cut off:]
(TODO: clarify method
1) find the good trustworthy judge [?] (mi: check)
2) find a good trustworthy Trust Index that replaces – hopefully – human evals;
how may we verify judges)

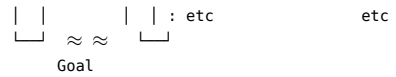
A.3 `37241.jpg` — notebook: "What trust index makes sense to design?"

Q: What trust index makes sense to design?
↪ like [what is expensive] in the "certification" of AI judges?

* Human labels P_θ(s | x, ŷ, y*, r) = aj(score | x, y*, ŷ, r)
↪ "ideal judge just … (ŷ, r)"
best case is AI Judge aj behaves exactly as the distribution of humans
↪ "ideal judge" depends only on ŷ, e.g. aj(score | ŷ, r) ↪ "rubric"

↪ so ideally P_θ(s | x, y*, ŷ, r) = P_humans(s | x, y*, ŷ, r)
some llm OW (Open Weights) or CW (Closed Weights)
all humans (or H = Humans), index

P_θ(s|x,y*,ŷ,r+i) ← Brando P_h(s | x, y*, ŷ, r+b)
s_θ~ ← Yegor P_h(s | x, y*, ŷ, r+g)
s_θ ← Rylan P_h(s | x, y*, ŷ, r+r)



→ What is a h in H? Could be just the name/index, or a description of the
personality of the person (or set / type / groups of people/humans)
↪ different types of guarantees can be made

∀h_j P_θ(s | x, y*, ŷ, r+i) ≈ P_H(s | x, y*, ŷ, r+h_j) (0)
∀aj = P_θ distribution equivalence

A.4 `37243.jpg` — annotated printout (Figs. 13–14)

[printed: "Certifying the Judge"; Figure 13 (a)(b); Figure 14 – see §3.2]

Goal: Given a benchmark that is difficult to score, how do we scale the
trustworthiness of flexible LLM judges (as the scoring fun / metric) for any
agent input (or at least the ones on the leaderboard)?

Q: do we really need to scale it? What does scaling mean here?
3 scaling params (I think):
∀a in A
∀b in B ^ AI Judges.
∀aj in AJ

Get [scribbled]_aj(ŷ, b, a) ≈_c h_j(ŷ, b, a)
↪ I think it should scale wrt new a agents & aj's ai judges?
↪ because if a new smarter ai judge aj comes – Clandeuloo or whatever – we want
it to still work. => but then, why don't you just use the human labels you had?

A.5 `36784.jpg` / `37240.jpg` — TODOs sheet

TODOS:
→ we need to explain the problem we are solving precisely (I think it is
[scribble] be-sc / bench-scalable-oversight); with time complexity arguments.

→ We need to argue why our method generalizes to all agents/files we try to judge
[scribble] or the most important ones, e.g. SOTA ones, at least for leaderboards.
↪ i.e. ∀a, TI(a, B) ≈_c HJ(a, B) (for all a in Agents)
↪ I think showing "test error" / having a val set or such is "low"
(statistically significant / practically significant) is a good idea?

↪ So: the argument: [struck out]
is TI ≈ Corr(J, Humans)
TI(J) ≈_c Corr(J, Humans)

=> So we can use TI to justify using a judge for … spec judgment for:
∀a in A & ∀b in B
↪ for all agents that could try to solve the task
↪ all benchmarks? Or just for VB
↪ I think just for VB.

A.6 `37242.jpg` — notebook: moments / stability ladder

or maybe you just want the "samples" to be the same, which is easier to do cuz
it's really saying the 1st moment match.

[struck out] ∀h_j in H, ∀aj in J,
E_{P_θ(s|x,y*,ŷ,r+aj)}[s] ≈ E_{P_H(s|x,y*,ŷ,r+h_j)}[s] (1)
↪ on avg the scores match ≈ sampling P_θ ≈ P_H (roughly)

"same variance/info" (2)
→ If we want [stability] we can require same variance:
(↪ so has same variance/variation as humans)
or low variance (3)
Stability.

```
∀h_j in H, ∀aj in J,  Var_{P_θ(s|x,ŷ,y*,r+aj)}[s]  ≈  low ≈ 0            (3)
  ( word didn't depend on humans lol… )
   or
∀h_j in H, ∀aj in J,  Var_{P_θ(s|x,ŷ,y*,r+aj)}[s]
                                ≈  Var_{P_H(s|x,y*,ŷ,r+h_j)}[s]       (2)

0 = ai / human judges behave the same as our judge
1 = ai judges behave on avg (sample) as human judge
2 = our ai judges behaves the same on avg (sample level) + same variation as humans
3 = our ai judges behaves the same on avg (sample level) and is stable
    (i.e. our judge is stable).
```

A.7 `37245.jpg` — wall-pinned sheet: the metric statement

```
We want ∀h_j in H, ∀aj in J,
   P_H(s | x, ŷ, y*, r+h_j)  ≈^{||.||^M}  P_θ(s | x, y*, ŷ, r+aj)
      ↪ s.t. h_j ≈_M aj

 ↪ for any aj, the probability of score s coming out is the same as the
   probability of any human (matching human & judge, of course, like the
   corresponding persona. Like, obviously if h_j = risk-taker persona but
   aj = conservative/shy, then the two cannot be made to match. lol).

||.||^M = whatever metric we want, e.g. KL-div / total info /
          total variation distance or whatever.
```

A.8 `37244.jpg` — loose sheet: scores, correlation, scatter

```
[scribbled] * we want all judges that match (aj ≈ h_j) to have / pred / corr /
  approx each other for every benchmark & every agent response.

Want ∀aj ≈ h_j, ∀b, ∀a,  P_θ(s | x, ŷ, y*, r+aj)  ≈  P_H(s | x, ŷ, y*, r+h_j)

   note: ŷ = a(x), like some sample from the judge (persona)
         y* = human label. But not indexed; fix according to benchmark b.
     ↪ h_j instead is the persona (rubric + persona index/description) of the
       judge, like P_H(s|…)  (not P_A(s|…), not notated at all).

   P_θ(s | x, ŷ = a(x), y*, r+aj)  ≈  P_H(s | x, ŷ = a(x), y*, r+h_j)

[left column]
   = s(x, ŷ, y*, r+aj)
   ŝ  ~ P_θ(s | x, ŷ, y*, r+aj)
   s* ~ P_H(s | x, ŷ, y*, r+ah)  ≜  s*(x, ŷ, y*, r+ah)

→ ok let's try something easier for now. Let's have the score approximate each
  other in some sense (e.g. correlate, predict, etc):
   ∀aj ≈ h_j, ∀b, ∀a,  ŝ(x, ŷ, y*, r+aj)  ≈  s*(x, ŷ, y*, r+h_j)
                                          ↪ same "sense" e.g. Correlate.

→ This can be argued [can be approximated by] the avg. score.
  (to remove the sampling; otherwise we'd need a guarantee in probability
   or something)
 ↪ actually can be done with scatter plots & correlations.

   ŝ(x,ŷ,y*,r+aj)
    1.0 ┤      x x
        │   x  x      rho = 0.87 etc, e.g. rho_spearman, rho_k, rho_pearson, (rho_ICC?)
        │  x x
    0.0 └──────────→  s*(x, ŷ, y*, r+h_j)

   Corr(ŝ, s*) ≜ Corr(ŝ ≈ s*) ≜ || ŝ ≈ s* ||^C
```

A.9 `37232.jpg` — desk context photo

No unique content. Establishes that A.3, A.4 and A.8 were worked on in one sitting, which is consistent with their shared notation.

Appendix B — Consolidated formal chain

(B0) Distributional equivalence, persona-matched — the definition, at M_0:

$$\forall (aj, h_j) : h_j \approx_{\Pi} aj,\ \forall b \in B,\ \forall a \in A,\ \forall (x, y^*) \in b : \left\| P_\theta(\cdot \mid x, y^*, a(x), aj) - P_H(\cdot \mid x, y^*, a(x), h_j) \right\|^M \leq \varepsilon$$

(B1) First-moment match: $\mathbb{E}_{P_\theta}[s] \approx \mathbb{E}_{P_H}[s]$

(B2) Second-moment match (fidelity to human disagreement): $\mathrm{Var}_{P_\theta}[s] \approx \mathrm{Var}_{P_H}[s]$

(B3) Stability, label-free: $\mathrm{Var}_{P_\theta}[s] \approx 0$

(B4) Score-level correlation surrogate: $\mathrm{Corr}(S_{aj}, S_{h_j}) = \text{high}$

(B5) Chosen matching criterion M_4: **(B4) and (B3)**

(B6) Label-free certificate: $\mathrm{PTI}_p(aj) = \mathrm{GM}(P_1(aj), P_2(aj), P_3(aj), P_4(aj))$

(B7) Bridge claim, validated once on held-out judges, $b = \mathrm{VB}$:

$$\mathrm{PTI}_p(aj) \ \approx_C\ \mathrm{TI}_{m_1}(aj), \qquad \mathrm{TI}_{m_1}(aj) = \mathrm{Corr}_{i \in I}\big(s_{aj}(i),\ \bar{s}_{H_{\approx_\Pi aj}}(i)\big)$$

(B8) Deployment: for any new judge aj' or new agent a', compute (B6) only. (B7) is never re-run.

The paper's job is (B7): show that the **label-free** quantity (B6) predicts the **label-requiring** quantity, on a held-out judge population, so that the right-hand side never has to be measured again.

Appendix C — Notation reconciliation log

The handwritten pages are internally inconsistent in a few places. Every choice made in these notes is recorded here so nothing is silently "fixed."

#	In the originals	Problem	Standardized to
1	$\hat{s}(x, \hat{s}, y^*, r{+}aj)$ — `37253`, `37244`, `37252`	$\hat{s}$ appears in two roles: the output score *and* the second conditioning slot	Second slot is $\hat{y} = a(x)$, the **artifact**. Confirmed by `38700`, which writes $P_\theta(s \mid x, y^*, \hat{y}{=}a(x), aj)$
2	Argument order: $(x, \hat{y}, y^*, \cdot)$ vs $(x, y^*, \hat{y}, \cdot)$	Both orders appear	$(x,\ y^*,\ \hat{y},\ \text{judge})$, following `38700`
3	Judge index written $r + aj$ vs aj	Rubric sometimes separate, sometimes folded in	Judge identity $= (r, \pi)$, written aj. Rubric is part of the judge, not a free argument
4	$\mathrm{Var}(\hat{s}, s^*)$ — `37253`	Variance is not a two-argument function	Judge's own repeat variance $\mathrm{Var}_{P_\theta}[s]$, i.e. the P_4 / stability term

#	In the originals	Problem	Standardized to
5	$\|\cdot\|^M$, $\|\cdot\|^C$, $\approx_{P_c}$, $\approx_c$, $\approx_M$	Five notations for "matches in sense X"	One matching criterion d_M, with $M \in \{M_0, \dots, M_4\}$ (§1.3)
6	Persona relation written $h_j \approx_M aj$ — `37245`; and $aj \approx h_j$ — `38700`	Overloads M, which is also the divergence	Persona relation is $\approx_\Pi$; divergence is d_M. **Distinct symbols**
7	$\mathcal{H}$, H, $H(b)$, "Humans"	Human population inconsistently named	$H(b)$ = human judges relevant to benchmark b
8	$\mathrm{TI} \approx \mathrm{Corr}(J, \mathrm{Humans})$ — `36784`	Type error: judge-indexed vs item-indexed	$\mathrm{TI}(aj) \approx_C \mathrm{TI}_{m_1}(aj)$, correlated **over judges** (§0.3, Alg. 5)
9	$\mathrm{GM}(\mathrm{Corr}(\hat{s}, s^*), 1 - \mathrm{std})$ — `37253`	Contains s^*, so the "certificate" is not label-free	$\mathrm{PTI}_p = \mathrm{GM}(P_1, P_2, P_3, P_4)$; the correlation term moves to the bridge (§3.3)
10	$\Delta 4$ / $P4$ label on the stability arrow — `37253`	Ambiguous glyph	Read as **P4 = stability**, consistent with the property list
11	Levels 0–3 (`37242`) vs options 0–2 (`38700`)	Two overlapping ladders	Merged into the single ladder $M_0 \dots M_4$ (§1.3)
12	"Coden" / "Clandeuloo"	Illegible proper nouns	Read as Codex-class agent and a placeholder future model name. Marked `[?]`
13	M_i subscript — `38700`, `38704`	One subscript carrying three jobs: TW definition, conditioning set, primed variant	Definition → superscript TW_k; conditioning set → written out in brackets; no method numbers (§1.5)
14	TW_i (`38704`) vs. options 0/1/2 (`38700`) vs. levels 0–3 (`37242`)	Three numberings of the same ladder	$\mathrm{TW}_2 \equiv$ option 2 $\equiv M_4$ (correlational + stable). Cross-reference table below
15	h_j used as a conditioning argument	No such pointer exists operationally	Method conditions on D_{h_j}, a finite score sample; h_j is latent (§1.6)
16	"person + persona ≈ persona" — flagged as an overload by the author	Individual identity conflated with described role	**Person** (individual, non-transferable) vs. **persona** π (described role, rubric-specifiable). $\approx_\Pi$ acts on π only (§1.6.1)
17	$\forall a \in A$ (Def. 1) vs. L_A (`38704` working target)	Definition quantifies wider than the experiment	Make the agent set an explicit parameter of Definition 1; claim the L_A version (§1.5.1)
18	$D_H^S = \bigcup_{h_j}\{s\}$ (`38707`)	Union discards the rater index	Indexed family $\{(h_j, x, \hat{y}, s)\}$ (§1.6.3)
19	$N_{h_j}^S$ vs "10 scores" (`38704`) vs "15 judgments" (`38707`)	Per-person count stated three ways	$N_{h_j}^S$ is the

per-person count; its realized value is ≈ 15, possibly as $10 + 5$ batches. Resolve from data |
20 | $D^S_{h_j}$ vs D_{h_j} | Superscript S used inconsistently | $D^S_{h_j}$ = the score set; the S marks scores, not a variant. Kept |

21 | "Trust Index" vs TW_k | One letter apart, opposite meanings (instrument vs target); and the name asserts the conclusion | Trust Index → **Proxy Trust Index**, $\mathrm{PTI}_p(aj)$. Never abbreviated `CI` (collides with confidence interval) (§0.4) |
22 | P_4 as "a property predicting trustworthiness" | Under TW_2 it is *literally half the definition*, not a proxy | $\mathrm{TW}_2 = [\mathrm{TI}_{m_1}$ high$] \wedge [P_4$ high$]$; the bridge only carries P_1, P_2, P_3 (§3.5) |

23 | `TI_p` (`20260812_1348`) | p indexes properties, which exist only on the proxy side | Read as TI_m; gold scalar is indexed by the matching criterion m |
24 | M_0–M_4 (these notes, capital) vs m_0–m_2 (`20260812_1348`, lowercase) | Two ladders | **Lowercase $m_0/m_1/m_2$ is canonical** (ratified). Mapping: $M_0 \to m_0$, $M_3 \to m_1$, $M_4 \to m_2$; M_1/M_2 (moment levels from `37242`) remain as unnumbered intermediate relaxations in the derivation |
25 | $\mathrm{TI}_{\mathrm{var}}$ / Cert / Proxy Trust Index | Three successive names for one object | PTI_p, final (§0.4). Align → TI_{m_1} |

26 | GPT-2nd-round: tilde/hat pair ($\widetilde{\mathrm{TI}}_m$ uncalibrated, $\widehat{\mathrm{TI}}_m$ calibrated) | Coherent, but the calibrated object does not exist at $n_{\mathrm{judges}} \approx 8$ | **Declined for now.** Revisit at $|AJ| \approx 40$ when a calibration map f_{cal} is fittable; the hat is earned by that map |
27 | GPT-2nd-round: rename $\mathrm{TI}_m \to \mathrm{Align}_m$ | "Align" severs the visible m-link between criterion and target; the construct/criterion gap is already carried by the role labels (§0.4) | **Declined.** TI_m stays; "Trust Index" banned from title/branding only (collision confirmed: arXiv:2507.16028) |
28 | "Certification protocol" (Part IV, Algs. 3–4) | Inconsistent with the retirement of certification claims | Renamed: **amortized judge validation**; CERTIFY → SCREEN; "certificate extends" → "screen result extends" |

29 | $(\mathrm{Corr} \star \mathrm{STD})^{1/2}$ (`39217`) | Missing orientation flip — rewards instability | $1 - \mathrm{STD}$; same $\Delta 4$ fix as `37253` (§1.7) |
30 | $\mathrm{STD}_{P_\theta}[D_{\hat{S}^*_H}]$ (`39217`) | P_θ is the judge; STD of human scores would break the label-free conjunct | $\mathrm{STD}[D_{\hat{S}_{aj}}]$ — stability is P_4, judge-side |
31 | $\forall x, y^*$ outside Corr (`39218`) | Pointwise syntax on a correlational statement | Items are integrated over, not quantified; residual choice is pooled vs per-agent Corr (§1.7) |
32 | $75 = 5 \times 15$ ratings (`38707`) vs 75 items $\times 2$ passes $= 150$ ratings (paper) | Mutually exclusive decompositions | **RESOLVED:** 5 *splits* $\times$ 15 rows, one rater; 150 rated / 144 analyzed (§1.8.2) |
33 | Table row "IS" (`39218`) | Matches none of the five VB subsets | `[?]` — still open; check repo subset configs |
34 | pass1/pass2 (paper) | Inverted vs. chronology; sha-verified | Read as strict/lenient; rename in paper + code (§1.8.3) |
35 | $|b|$ ambiguity | 157 (problems) vs 857 (theorems) | Two granularities, both real; state which each count uses |
36 | `39218` agent columns CC/Codex/Leanstral/Qwen | Mixed agents with judges | Annotated agents = codex/claude/llama; Leanstral/Qwen are judge families |
37 | $|b|$ in gold-side tensors (`39296`) | Overstated coverage | $b' \subseteq b$ on every gold-side D; PTI alone runs on full b (§1.8.1) |
38 | $\mathrm{TW}_{2'}(b, aj, L_A, H) = (\mathrm{Corr} \cdot \mathrm{STD})^{1/2}$ (`39399`/`39398`) | Assigns a scalar formula to the predicate symbol | Read as $\mathrm{TI}_{2'}$; $\mathrm{TW}_m := [\mathrm{TI}_m \geq \tau]$ |
39 | Missing orientation flip, **third occurrence** (`37253`, `39217`, `39399`/`39398`) | Raw STD rewards instability | $1 - \mathrm{STD}$; add to the manuscript check-list |
40 | $\hat{S}^*_{A|b,L_A}$ in the goal box (`39398`) | A is the agent set — type error | Read as H |
41 | $L_A \subseteq L_A$ (`39398` margin) | Same symbol both sides | Read as $L'_A \subseteq L_A$ (leaderboard subset actually run) |
42 | $|\mathrm{mb}(\cdot)|$ orientation (`39404`) | Absolute value also rewards $\rho \to +1$ (anti-monotone judge) | $\frac{1-\rho}{2}$, matching the paper's C_f map |

43 | $\mathrm{TI}_{2'}(b, ...)$ in the boxed target (`39404`) | Gold is only measurable on b'; the page's own note says "b is whole b not b'!!" for PTI | $\mathrm{TI}_{2'}(b', ...)$; the b vs b' asymmetry is the amortization |
44 | $\approx_m$ used for both the item-level criterion and the PTI–TI bridge (`39404`) | Two different index sets one level apart | Bridge gets $\approx_C^{AJ}$ (over judge configurations) |
45 | $P_{\text{ps\&SI}}$ (`39403`) | Wording broader than the experiments | **RESOLVED (`39658`):** fourth property is $P_{\alpha SI}$ (α-semantic invariance); repeat/prompt variance stays in the target's $(1 - \text{std})$ conjunct. $K = 4$ |
46 | $\prod_{y \in D}(\cdot)$ raw products (`39658`, `39657`) | Scale-dependent: value depends on $\lvert D \rvert$, so PTI on b and TI on b' become incomparable by construction | Geometric mean $(\prod)^{1/\lvert D \rvert}$ at both aggregation levels |
47 | $1 - \rho$ (`39657`) | Range $[0, 2]$; aggregate can exceed 1 | $\frac{1-\rho}{2}$ |
48 | $O(\lvert H \rvert \cdot \lvert b' \rvert)$ (`39659`) | Missing the per-problem candidate factor | $O(\lvert H \rvert \cdot \lvert b' \rvert \cdot \lvert L_A \rvert)$; reproduces the audited 144 |
49 | bare std (`39658`) vs $1 - \text{std}$ (`39659`) | Same page-pair disagrees | `39659` is correct; adopt. Confirmed again in blue on `39970` |
51 | $\mathcal{M}^{VS}$ used verbatim for both target and properties (`PXL_20260817`) | Shared $(1 - \text{std}(\widehat{S}_{aj}))^{1/2}$ factor makes part of the bridge correlation mechanical | Properties use $\mathcal{M}^{C}$ (correlation only); target keeps $\mathcal{M}^{VS}$. Report $\mathrm{Corr}(\sigma_{\text{target}}, \sigma_{\text{proxy}})$ as a diagnostic |
52 | Semicolon convention (`39969`, `PXL_20260817`) | Consistent across pages | Adopted: varying args before `;`, fixed context after. Notes may use \| for legibility; translate to `;` in the manuscript |
53 | $1 + \rho$ in the boxed P_{mb}^{GM} (`PXL_20260817_1938`) | Inverted: ideal $\rho = -1$ maps to 0 | $\frac{1-\rho}{2}$ |
54 | "std" inside a ladder property | Ambiguous between spread along the ladder (should be large) and repeat variance at a rung (should be small) | Repeat variance at fixed rung, averaged across rungs |
55 | Variance placement | Earlier recommendation was Config A (properties correlation-only) | **Superseded: Config B** — gold is plain human correlation, properties carry variance; matches the existing implementation |
56 | "high Corr/VS for all y" applied to α-renaming | Correlation undefined against a constant second argument | Invariance properties take the deviation form $1 - \mathbb{E}\lvert \Delta s \rvert$ |
50 | Anchor judges in the bridge scatter (`39969`) | Random and good-judge anchors span nearly the whole unit square | Keep them for scale legibility, but report correlations **primaries-only and pooled**; controls can carry the effect |

Ladder cross-reference. The numberings, aligned (rightmost column canonical):

`37242` level	`38700` option	`38704` name	Old (these notes)	**Canonical**	Condition
0	0	TW_0	M_0	m_0	Distributional equality
1	—	—	M_1	— (relaxation)	First moment
2	—	—	M_2	— (relaxation)	First + second moment (matched to humans)
—	1	TW_1	M_3	m_1	Correlational ("most common")
3	2	TW_2	M_4	m_2	**Correlational + stable ("fav") ← the working target**